



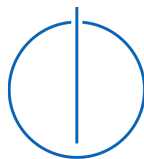
SCHOOL OF COMPUTATION, INFORMATION
AND TECHNOLOGY

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

**ECHR-QA: Designing a Question Answering
Benchmark for European Court of Human
Rights Case Law**

Korbinian Weidinger





SCHOOL OF COMPUTATION, INFORMATION
AND TECHNOLOGY

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

**ECHR-QA: Designing a Question Answering
Benchmark for European Court of Human
Rights Case Law**

**ECHR-QA: Entwurf einer Frage Antwort
Benchmark für Rechtsprechungen des
Europäischen Gerichtshofs für
Menschenrechte**

Author:	Korbinian Weidinger
Supervisor:	Prof. Dr. Matthias Grabmair
Advisor:	Yaswanth Tokala
Submission Date:	September 22, 2024



I confirm that this thesis is my own work and I have documented all sources and material used.

Munich, September 22, 2024


Korbinian Weidinger

Abstract

Large Language Models (LLMs) are increasingly regarded as transformative in the legal industry, with the potential to significantly reduce costs and enhance efficiency through the automation of routine legal tasks. However, the responses they generate frequently contain errors, biases, hallucinations, and other issues that could mislead professionals and lead to societal harm. To improve the trustworthiness of LLMs, one promising approach is to attribute answers to actual sources, thereby increasing the factuality and verifiability of the information provided. We introduce **ECHR-QA**, a **E**uropean **C**onvention on **H**uman **R**ights **Q**uestion-**A**nswering Benchmark, which consists of information-seeking questions derived from ECHR jurisprudence, paired with attributions to relevant judgments, and is specifically designed to advance attributed legal question answering. This initiative directly addresses the critical need to enhance the correctness and verifiability of responses generated by LLMs in the legal domain, tackling prevalent issues such as errors, biases, and misinformation. To efficiently curate this dataset, we developed specialized strategies in collaboration with an expert ECHR lawyer to automatically extract question-answer pairs from ECHR case-law guides. To establish a high-quality, attributed question-answering benchmark, we devised an LLM-based filtering pipeline aimed at enhancing dataset quality, utilizing expert annotations as a benchmark to evaluate the efficacy of QA pairs. While recent applications of LLMs as evaluators have demonstrated promising results, our filtering approach has only achieved limited success, likely due to the inherent complexity of legal texts coupled with the current limitations of LLMs. We evaluated several LLMs on our established benchmark to highlight the significant challenges that current models face in attributed legal question answering, including one specifically trained on legal corpora. Our comprehensive evaluation highlights the persistent difficulties in achieving accuracy and reliability, emphasizing the urgent need for enhanced modeling strategies to improve the performance of LLMs in handling the nuanced and complex nature of legal environments. Conclusively, this thesis underscores both the difficulties and the potential in automating the development of a legal QA benchmark. Our efforts aim to contribute towards the establishment of more reliable and accurate legal AI applications, which are essential for supporting the professional and ethical deployment of AI technologies in high-stakes fields such as law.

Acknowledgments

I want to take this opportunity to express my sincere gratitude to everyone who contributed to the completion of this thesis:

First, I must thank my supervisor, Yaswanth Tokala. His oversight extended beyond just supervising this thesis; he also led the master practical course “Legal Data Analysis Lab” which was an incredibly enriching part of my academic journey. Without his guidance, persistent support, and dedicated involvement, this thesis would not have been possible.

I also wish to express my gratitude towards Prof. Matthias Grabmair. His lectures not only deepened my interest in legal tech but also provided significant support during my thesis writing process. His course, along with the “Legal Data Analysis Lab” rank among my most rewarding academic experiences.

My biggest appreciation is reserved for Oana Ichim, whose contributions were indispensable to this thesis. She devoted countless hours to this work, providing detailed explanations and valuable insights, and showed remarkable commitment and motivation. Without her, this thesis could not exist. Oana was more than just a collaborator; she also became an advisor and a reliable support through the many challenges of this work. She helped not only with the academic challenges of this thesis but also offered guidance on broader issues which I greatly appreciate.

My thanks also extend to Dr. Dominic Henze and Dr. Matthias Grundmann, who have become mentors and friends over the last few months and with whom I thoroughly enjoy working.

Additionally, I am thankful to my father, who introduced me to coding and has always enthusiastically supported my academic endeavors. I am equally grateful to my mother, whose advice has been invaluable not just for this thesis but throughout my life, and who has always been there to listen and support. I must also thank my cousin Mathias, who has been both a close friend and a mentor during my studies at TUM. Lastly, my gratitude extends to my girlfriend, who supported me through my master’s degree and helped me navigate the tougher times with her consistently positive attitude.

Each of these individuals has profoundly impacted my educational journey and the completion of this thesis, and I am deeply grateful for their support and encouragement.

Contents

Abstract	iii
Acknowledgments	iv
1 Introduction	1
1.1 Background and Motivation	1
1.2 Approach and Goals	1
1.3 Structure of this Work	2
2 Related Work	4
2.1 Hallucinations	4
2.2 QA Datasets and Benchmarks	5
2.3 Attributed QA Datasets and Benchmarks	5
2.4 LLMs as Evaluators	6
2.5 Prompting Techniques	6
2.6 Legal Question Answering	7
2.7 ECHR Related Work	7
3 ECHR-QA Dataset Generation	8
3.1 ECHR-QA Task	8
3.2 Case-law Guides	8
3.3 QA Pair Curation Strategies	9
3.4 QA Pair Evaluation	10
3.4.1 Evaluation Metrics	11
3.4.2 Expert Evaluation Results	11
3.4.3 Automatic Evaluation	12
3.5 ECHR-QA Dataset	13
4 Attributed QA Approaches	15
4.1 Retrieval	15
4.2 Models	15
4.3 Approaches	16
5 Evaluation	18
5.1 Automatic Evaluation	18

Contents

5.2 Results	20
5.3 Expert Evaluation	23
6 Conclusion	27
7 Limitations	28
8 Ethics	32
9 Future Work	33
A Appendix	35
A.1 ECHR-QA Dataset Case Study	35
A.1.1 Example QA Pairs	38
A.2 Attributed QA Case Study	43
A.2.1 Example Generations	46
A.3 BM25	50
A.4 Figures	51
A.5 ECHR-QA QA Pair Curation Prompts	53
A.6 ECHR-QA QA Pair Curation Code	59
A.7 Attributed QA Prompts	60
A.8 Annotation Platform	68
Acronyms	72
Bibliography	73

1 Introduction

1.1 Background and Motivation

Recent advancements in LLMs - models trained on large amounts of textual data to predict the next token in a sequence - significantly impact the legal landscape (Fagan, 2024; Frankenreiter et al., 2022; Martin et al., 2024). These applications impact various legal areas, from legal education (Choi et al., 2023; Katz et al., 2024), to legal research (Livermore et al., 2023), to law firms and legal technology start-ups incorporating LLM-based tools to enhance operational efficiency and reduce costs, making legal information and services more accessible and affordable to all (Fagan, 2024; Rodgers et al., 2023). LLMs are increasingly performing better at law-related tasks, such as the bar exam (Katz et al., 2024), statutory reasoning (Blair-Stanek et al., 2023), and Issue-Rule-Application-Conclusion (IRAC) analysis (Guha et al., 2024). However, the responses they generate frequently contain errors, biases, and hallucinations, resulting in outputs that may not be consistent with prevailing case law and legal doctrines which could mislead professionals and lead to societal harm (Dahl et al., 2024). This shortcoming continues to hinder the broader adoption of these tools in the legal profession (Dahl et al., 2024). Recent incidents underline the risks associated with lawyers relying on LLMs. A lawyer used ChatGPT for legal research and submitted court filings containing fabricated quotes from nonexistent judicial decisions¹ and other lawyers created legal briefs full of made-up cases and citations².

To reduce the risk of hallucinations, enable verifiability, and increase transparency, it is essential to develop models that incorporate genuine citations to supporting evidence (Bohnet et al., 2022; Kamalloo et al., 2023). This thesis aims to contribute towards the development of more trustworthy, attributed AI systems, for the legal domain, by providing an attributed question-answering (QA) benchmark dataset for the European Convention on Human Rights (ECHR) case law.

1.2 Approach and Goals

To investigate the ability of current and future LLMs to generate responses with citations for legal queries, the creation of high-quality benchmark datasets is imperative. This work proposes a strategy for creating such QA benchmark datasets for the legal domain at scale and

¹<https://www.nytimes.com/2023/05/27/nyregion/avianca-airline-lawsuit-chatgpt.html>

²<https://www.nytimes.com/2023/06/22/nyregion/lawyers-chatgpt-schwartz-loduca.html>

outlines a framework for the automatic evaluation of LLM performance on said benchmarks. Our first goal is to develop a high-quality attributed question-answering benchmark dataset based on ECHR case law documents (ECHR-QA), employing LLMs and an ECHR expert for quality assurance. ECHR-QA is built from the ECHR case-law guides³, which cover various legal concepts relevant to each article of the convention by referencing specific paragraphs from ECHR judgments. We develop various strategies that use LLMs to extract responses with citations from these guides as well as generate questions. This approach is inspired by recent advancements in using LLMs for instruction dataset creation (Köksal et al., 2024; X. Li, Cao, et al., 2023; Ushio et al., 2023). We validate the effectiveness of the QA pair generation strategies by collaborating with an ECHR expert, assessing them across multiple dimensions including fluency and comprehensiveness. Based on previous work using LLMs as evaluators (Fu et al., 2023; Y. Liu et al., 2023; Pan et al., 2024; J. Wang et al., 2023; Zheng et al., 2024), we attempt to filter our results by devising an automated low-quality data filtering strategy. After the creation of the ECHR-QA dataset, we assess different attributed QA strategies including RAG (Lewis et al., 2020), LLatriveal (X. Li, Zhu, et al., 2023), Post-hoc, and RARR (L. Gao et al., 2022) with a range of state-of-the-art LLMs. We then introduce an automatic evaluation framework for the ECHR-QA dataset which assesses the generated responses and corresponding attributions across answer fluency, answer correctness, citation faithfulness, and citation relevance. Lastly, we compare the automatic evaluation of the results with rankings from our ECHR legal expert to assess the effectiveness of the framework. We aim for our benchmark, ECHR-QA, to benefit the legal NLP community by facilitating the development of enhanced methods for long-form legal question answering and the creation of automated metrics for their assessment.

1.3 Structure of this Work

In 1 Introduction we describe the goals of this thesis and motivate why we create an attributed legal long-form question-answering benchmark dataset for ECHR case law.

We continue with chapter 2 Related Work which additionally provides relevant background knowledge to our work. It includes general prior work on hallucinations (2.1 Hallucinations) in the legal domain, information about prior question-answering benchmarks (2.3 Attributed QA Datasets and Benchmarks), how LLMs can be used as evaluators (2.4 LLMs as Evaluators), introduce current trends in prompting techniques (2.5 Prompting Techniques), previous work in legal question answering (2.6 Legal Question Answering) and other ECHR related work (2.7 ECHR Related Work).

Chapter 3 ECHR-QA Dataset Generation walks the reader through the journey of the ECHR-QA dataset curation. We start by introducing our set of explored QA pair curation strategies (3.3 QA Pair Curation Strategies). The resulting QA pairs were evaluated by an ECHR expert to get a better understanding of the benefits and drawbacks of our QA pair curation approaches (3.4 QA Pair Evaluation). We then describe how one can use expert annotations to build an LLM-based evaluator for a more scalable automated evaluation (3.4.3 Automatic Evaluation).

³<https://ks.echr.coe.int/web/echr-ks/all-case-law-guides>

We finish the chapter with the introduction of our ECHR-QA dataset ([3.4.3 ECHR-QA Dataset](#)). We include a case study with additional legal details regarding the curated QA pairs in the Appendix ([A.1 ECHR-QA Dataset Case Study](#)) highlighting key dataset properties with the help of our ECHR legal expert.

After introducing our attributed long-form QA dataset ECHR-QA we present a set of question-answering approaches including different RAG and Post-hoc attribution strategies executed with different state-of-the-art LLMs ([4 Attributed QA Approaches](#)).

To turn the ECHR-QA dataset into a benchmark, we present our automatic evaluation framework ([5.1 Automatic Evaluation](#)) in Chapter [5 Evaluation](#). We then present the evaluation results for the approaches and models introduced in Chapter [4](#) in section [5.2 Results](#). To showcase the effectiveness of the automatic evaluation we compare it to expert annotations ([5.3 Expert Evaluation](#)). We provide a legal analysis conducted in collaboration with our ECHR expert highlighting key differences between the implemented models and strategies ([A.2 Attributed QA Case Study](#)).

In Chapter [6 Conclusion](#) we summarize the findings of our work. We then discuss internal, external, and structural limitations of our approach ([7 Limitations](#)). We conclude this thesis with propositions for future work ([9 Future Work](#)).

2 Related Work

2.1 Hallucinations

Hallucinations - responses that contain either incorrect information or false assertions that a source supports a proposition - have been widely recognized as a significant drawback of LLMs. [Magesh et al., 2024](#) demonstrate that proprietary AI research tools, such as LexisNexis (Lexis + AI) and Thomson Reuters (Westlaw AI-Assisted Research and Ask Practical Law AI), hallucinate between 17% and 33% of the time, which is significantly lower than general-purpose chatbots like GPT-4, which have been found to hallucinate in up to and over 50% of cases, particularly on legal and other domain-specific knowledge-based questions ([Dahl et al., 2024](#); [Zuccon et al., 2023](#)). They additionally propose a clear typology for differentiating between hallucinations and accurate legal responses, including terminology for different types of hallucinations. Responses that contain factual inaccuracies are labeled as *incorrect*, which is a type of hallucination. On the other hand, *misgroundedness* refers to another kind of hallucination where the response cites relevant sources, but these sources do not support the claims made in the response. Similarly, [Rashkin et al., 2023](#) defines a statement as attributable to the source (AIS) if the given source entails it. Correctness and groundedness are vital for LLMs to do case law, establishing a framework of precedents that guide judicial decisions and legal reasoning. Proprietary tools, as well as recent research, have shown that retrieval augmented generation (RAG) and other techniques can reduce hallucinations ([L. Gao et al., 2022](#); [X. Li, Zhu, et al., 2023](#); [Magesh et al., 2024](#); [Shuster et al., 2021](#)). [Z. Xu et al., 2024](#) highlight that LLMs cannot learn all computable functions and will therefore always hallucinate, exploring existing hallucination mitigators as well as the practical implications on the safe deployment of LLMs. While the more promising approaches explore RAG techniques, other lines of work explore post-hoc attribution which generates a response first and retrieves evidence as attributions later ([L. Gao et al., 2022](#); [He et al., 2022](#); [Huo et al., 2023](#)). [L. Gao et al., 2022](#) propose a post-editing approach (RARR) where responses are additionally edited after retrieval. We investigate and contrast the RAG and post-hoc approaches within the ECHR-QA benchmark. While a lot of prior work extensively evaluates attributions with human evaluators ([Bohnet et al., 2022](#); [N. F. Liu et al., 2023](#); [Muller et al., 2023](#); [Nakano et al., 2021](#)), and ROUGE or BERT scores, recent work has proposed an automatic AIS metric (AutoAIS) ([Bohnet et al., 2022](#); [L. Gao et al., 2022](#)) by tackling it as an Natural Language Inference (NLI) task between the generated response and the attributed source using TRUE ([Honovich et al., 2022](#)). Others have explored automatic evaluation of attribution by prompting LLMs and

fine-tuning smaller LMs (Y. Li et al., 2024; Yue et al., 2023).

2.2 QA Datasets and Benchmarks

Early datasets, before the widespread adoption of LLMs, such as TriviaQA (Joshi et al., 2017), SQuAD (Rajpurkar et al., 2018), HotpotQA (Yang et al., 2018), Natural Questions (NQ) (Kwiatkowski et al., 2019), and ELI5 (Fan et al., 2019) used sources like Wikipedia or Reddit in combination with human annotators to curate QA datasets and used evaluation metrics like exact match (EM), F1, and ROUGE or short-answers for performance evaluation. ASQA (Stelmakh et al., 2022), focused on the lack of reliable metrics for measuring performance in long-form question-answering (LFQA) and dealing with ambiguous questions, highlighting the drawbacks of ROUGE measurements and proposing new measurements to capture completeness and comprehensiveness. Kamalloo et al., 2023 used human-machine collaboration to create HAGRID, which is based on the question and attribution passages from the existing information retrieval (IR) dataset, MIRACL (X. Zhang et al., 2023); they used LLMs to generate answers and human annotators to evaluate them on informativeness and attributability. While some datasets like HAGRID focus on golden answer generation, others also deal with the task of question curation. X. Li, Cao, et al., 2023 propose an evolutionary question generation strategy to control the question complexity and necessary knowledge of the answer.

2.3 Attributed QA Datasets and Benchmarks

Nakano et al., 2021 introduced WebGPT for which they used ELI5 to train GPT-3, allowing the model to search and navigate the web, generating answers supported by references collected during browsing. The performance of WebGPT was assessed using human evaluators, who determined that its answers surpassed those from the ELI5 dataset 69% and human-generated responses 59% of the time. Following WebGPT, Qin et al., 2023 introduced WebCPM, the first Chinese LFQA dataset. While WebGPT aims to mimic the behavior of a human by browsing the Web to answer a specific question, WebBrain uses a dataset for evidence retrieval. While the above works use structured documents like web pages as attribution sources, there have also been efforts to leverage structured knowledge bases instead (Amouyal et al., 2022; X. Li, Cao, et al., 2023). Amouyal et al., 2022 introduced QAMPARI, an attributed QA benchmark consisting of questions with multiple answers and supporting evidence, generated from Wikipedia’s knowledge graph which was manually paraphrased and validated.

Bohnet et al., 2022 study Attributed QA, investigating how to measure attribution and how well state-of-the-art methods perform on attribution using ELI5. They propose using a reproducible evaluation framework, AutoAIS (L. Gao et al., 2022), for the task and show that it correlates with human AIS annotations. Similarly, T. Gao et al., 2023 construct ALCE, a benchmark for automatically evaluating LLM generations across the three dimensions - fluency, correctness, and citation quality - and demonstrate their correlation with human judgments on ELI5, QAMPARI, and ASQA. Both Bohnet et al., 2022 and T. Gao et al., 2023 experiment with different retrieval strategies including retrieve-then-generate and closed-book generation with post-hoc attribution.

While most of the works deal with general knowledge based on Wikipedia or Reddit, we explicitly focus on constructing an attributed QA dataset for the legal domain, specifically the ECHR. We use human-machine collaboration to curate questions as well as golden answers with citations from ECHR case-law guides. Our research is therefore more comparable to a benchmark like EXPERTQA (Malaviya et al., 2023), which comprises information-seeking questions curated by experts across 32 fields, including law. However, it is important to highlight that legal principles are jurisdiction-specific and not universally applicable, in contrast to fields like medicine, which are more standardized globally.

2.4 LLMs as Evaluators

Creating a consistent method to evaluate LLMs in various contexts remains a significant challenge (Chern et al., 2024). Contemporary evaluation techniques frequently employ LLMs themselves to analyze the responses they generate (Fu et al., 2023; X. Li, Zhang, et al., 2023; Y. Liu et al., 2023; J. Wang et al., 2023; Zheng et al., 2024). Using LLMs as evaluators provides a scalable and explainable way to approximate human preferences, which are otherwise very expensive to obtain (Zheng et al., 2024). These LLM-based evaluations are typically meta-evaluated by determining the correlation between automated scores and human judgments using Spearman (Spearman, 1961) or Pearson (Mukaka, 2012) correlations, as discussed in recent studies like Fu et al., 2023.

2.5 Prompting Techniques

A prompt is the input provided to an LLM, which is used to direct its output. There are many different strategies in LLM prompting including zero-shot, one-shot, or few-shot prompts which describe how many demonstrations are provided at inference time (Brown, 2020). Another widely used prompting technique to improve LLM’s reasoning capabilities is Chain-of-thought (CoT) (Wei et al., 2022). CoT can be implemented with both zero-shot and few-shot prompting techniques (Gramopadhye et al., 2024; Liang et al., 2023; Zhao et al., 2023). Different prompts and prompting strategies are consistently being tested and evolved. Performance generally improves as techniques grow more complex, meaning few-shot or few-shot-CoT usually do better than zero-shot techniques (Schulhoff et al., 2024). This, however, can vary depending on the task and setting.

For example, the LLM evaluator, GPTScore, introduced by Fu et al., 2023, is originally based on a zero-shot framework but can also be adapted for few-shot settings. Similarly, our LLM-based evaluator introduced in Chapter 3 for QA pair filtering, uses zero-shot but could be adapted to use a few-shot prompt instead. We explore multiple attribution generation strategies on ECHR-QA, including RARR (L. Gao et al., 2022), which primarily utilizes few-shot prompting, and LLattribution (X. Li, Zhu, et al., 2023), which operates using zero-shot-CoT prompts.

2.6 Legal Question Answering

Legal Question Answering (LQA) ([Fawei et al., 2019](#); [Morimoto et al., 2017](#)) refers to the task of responding to legal questions. Answering questions within the legal domain is challenging primarily because of the complex and varied nature of legal document systems ([Abdallah et al., 2023](#)). This is usually done by a lawyer or another legal professional with expertise and knowledge in the relevant area of law. Legal question answering may entail several activities, such as researching existing laws, interpreting legal statutes and regulations, and applying legal principles and precedents to specific factual scenarios. LQA seeks to deliver precise and dependable information and advice on legal matters, assisting individuals and businesses in navigating the legal system and resolving legal issues ([Abdallah et al., 2023](#)). Various LQA datasets have been developed based on different legal sources, including on privacy policies ([Ahmad et al., 2020](#); [Ravichander et al., 2019](#)), regulatory documents ([Trivedi, 2018](#)), Belgian statutes ([Louis et al., 2021, 2024](#)), Vietnamese legal advice ([Bach et al., 2017](#); [Kien et al., 2020](#)), the Japanese bar exam ([Kim et al., 2015](#)), Chinese judicial exams and legal advice ([Chen et al., 2023](#); [Duan et al., 2019](#); [Huang et al., 2020](#); [Zhong et al., 2020](#)), contract reviews ([Hendrycks et al., 2021](#); [S. H. Wang et al., 2023](#)), German legal advice ([Büttner et al., 2024](#); [Hoppe et al., 2021](#)), and international private law ([Sovrano et al., 2021](#)).

2.7 ECHR Related Work

Previous work involving the European Court of Human Rights (ECtHR) has addressed a variety of tasks, including judgment prediction ([Aletras et al., 2016](#); [Chalkidis et al., 2019, 2021](#); [Santosh, Ichim, et al., 2023](#); [Santosh, San Blas, et al., 2023](#); [Santosh et al., 2022](#); [Valvoda et al., 2023](#); [S. Xu, Ichim, et al., 2023](#)), argument mining ([Habernal et al., 2023](#); [Mochales et al., 2008](#); [Poudyal et al., 2019, 2020](#)), event extraction ([Filtz et al., 2020](#); [Navas-Loro et al., 2022](#)), summarization ([Santosh, Aly, et al., 2024](#)), query-based relevant paragraph retrieval ([Santosh, Hernandez, et al., 2024](#)), vulnerability identification ([S. Xu, Staufer, et al., 2023](#)), and prior case retrieval ([Santosh, Haddad, et al., 2024](#)).

To the best of our knowledge, our work is the first ECHR long-form question-answering benchmark. We offer this benchmark to the research community to facilitate further advancements and research in the area of attributed (legal) question-answering and legal answer evaluation.

3 ECHR-QA Dataset Generation

This chapter details the process of creating the ECHR-QA dataset. Additionally, we present a case study on the ECHR-QA dataset, conducted by our expert in ECHR jurisprudence. All code used in the dataset generation is available on GitHub¹.

3.1 ECHR-QA Task

We characterize the task of attributable legal question answering as follows: Given a legal query q and a corpus of citeable text passages C obtained from the ECHR judgments, the system is required to generate an output text A that answers the question. The output text consists of a list of n sentences $\{a_1, \dots, a_n\}$, and each statement a_i can be followed by an in-text citation to a list of passages $C_i = \{c_{i,1}, c_{i,2}, \dots\}$, where $c_{i,j} \in C$. It should be noted that certain citation styles, such as "According to $[c_{i,j}]$, ..." are not explicitly covered by this formulation. However, these sentences can be rewritten to adhere to the specified format.

3.2 Case-law Guides

We curate our attributed QA dataset using the case-law guides² published by the ECHR, which are accessible on their knowledge-sharing platform maintained by the court's registry. These guides are designed to give legal practitioners a thorough understanding of the court's pivotal judgments and decisions. They provide a comprehensive analysis of how case law has evolved for each article of the European Convention on Human Rights and its additional Protocols as well as cross-cutting themes such as Data Protection and Rights of LGBTI persons.

In total, the guides cover 29 article-specific and 8 theme-specific topics, offering detailed discussions that include references to the court's judgments and decisions. We extract potential answers from these guides, aiming to generate attributed QA pairs. Rather than relying on costly and time-consuming legal expert annotations to produce gold-standard questions, we employ an off-the-shelf LLM, leveraging its generative capabilities to generate the questions.

¹<https://github.com/KorbiQWeidinger/echr-qa-dataset-generation>

²<https://ks.echr.coe.int/web/echr-ks/all-case-law-guides>

3.3 QA Pair Curation Strategies

The strategy for the creation of our QA pairs was developed over multiple stages in collaboration with our ECHR legal expert. All our strategies follow a chain-of-thought (CoT) approach (Wei et al., 2022). This section outlines the details of each strategy and provides references to the corresponding prompts. The development motivations and the insights gained about each strategy from our expert evaluation are detailed in Section 3.4.2 Expert Evaluation Results. We used GPT-3.5 (gpt-3.5-turbo-16k³) throughout the dataset curation process.

Single-Paragraph: We generate a question based on a single paragraph from the case law guides, with the provided paragraph serving as the answer to the generated question (Prompt: A.3).

Multiple-Paragraphs: We generate a question based on 5 sequentially selected paragraphs from a case-law guide, with the LLM selecting paragraphs serving as the answer to the generated question (Prompt: A.4).

Sentence-Level-CoT: We generate a question based on 5 sequentially selected paragraphs from a case-law guide, allowing the LLM to choose which sentences from the presented paragraphs to use as the answer (Prompt: A.5).

Sentence-Level-CoT with Search: We first generate a question given 3 paragraphs, selected from a case-law guide sequentially (Prompt A.6). Then we use the generated question to perform a semantic similarity search using embeddings (text-embedding-3-small⁴) across all the paragraphs in the respective case-law guide. This aims to identify ($k = 5$) paragraphs in the case-law guide that deal with the same information sought in the question, increasing the chance for the curated answer to be comprehensive. We then task the LLM to generate an answer with sentence-level citations from the initially presented and the additionally identified paragraphs (Prompt: A.7). We use these citations referencing case-law guide sentences to form our final answer. The goal of intermediary generation is to improve the fluency and argument flow of the answer by avoiding the strictly ordered sentence selection observed in the *Sentence-Level-CoT* approach. This ensures the final answer is fully extracted from the case-law guides preventing the risk of hallucinations. The process for *Sentence-Level-CoT with Search* is illustrated in Figure 3.1.

Legal Sentence-Level-CoT with Search: We then collaborated with our ECHR legal expert to formulate prompts to improve generated questions to improve our *Sentence-Level-CoT with Search* strategy even further. These efforts resulted in two new question-generation prompts: A.8 for *Legal Sentence-Level-CoT with Search v1* and A.9 for *Legal Sentence-Level-CoT with Search v2*, both of which use a redefined prompt A.10 to generate better answers

³<https://platform.openai.com/docs/models/gpt-3-5-turbo>

⁴<https://platform.openai.com/docs/guides/embeddings/embedding-models>

with sentence-level citations. The goal of the new question-generation prompts is to generate questions that avoid referencing specific cases, focusing instead on linking fact patterns to relevant legal doctrines, thereby creating more effective general information-seeking scenarios.

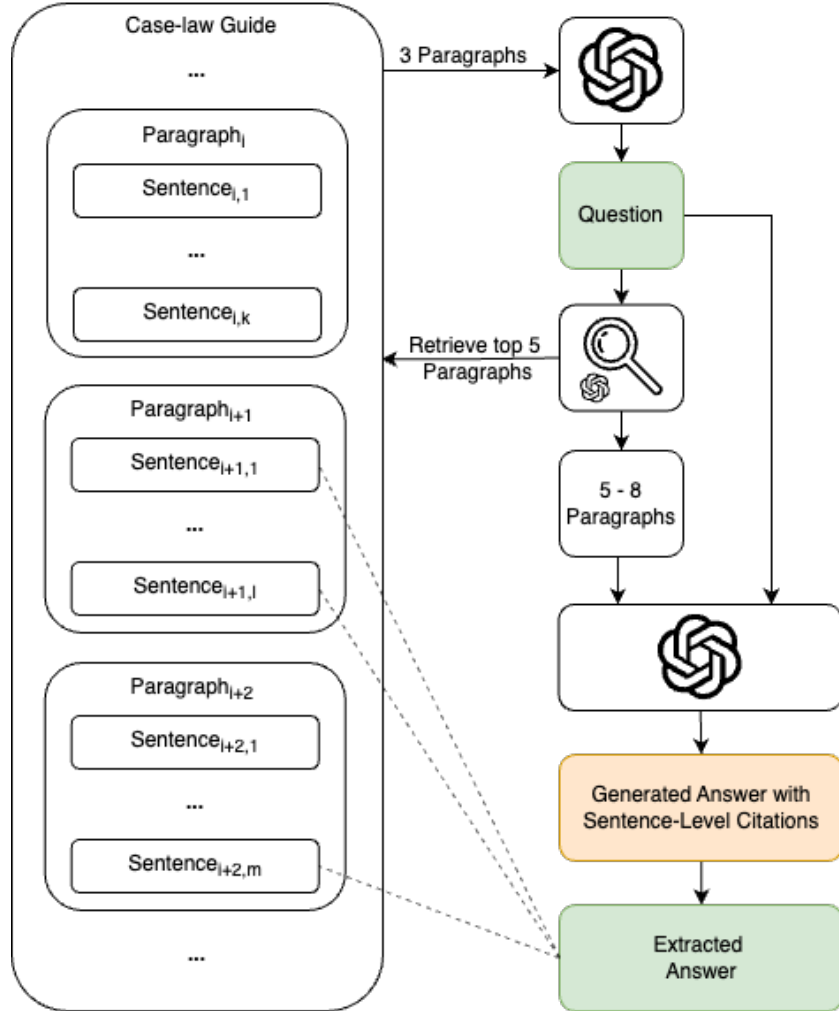


Figure 3.1.: (Legal) CoT with Search

All these strategies' final answers are extracted from the case-law guides. We do not use any generated answers to prevent the risk of hallucinations. For more details on the prompts see [A.5 ECHR-QA QA Pair Curation Prompts](#).

3.4 QA Pair Evaluation

Our legal expert evaluated 50 curated QA pairs for each strategy and wrote down the shortcomings of individual QA pairs and strategies.

3.4.1 Evaluation Metrics

Inspired by metrics proposed by other QA benchmarks (Funkquist et al., 2022; Kamaloo et al., 2023; Malaviya et al., 2023), we task our ECHR legal expert to annotate each QA pair on a scale of 1-5 across five dimensions:

- (i) Question Fluency: Is the generated question fluent and grammatically sound?
- (ii) Answer Fluency: Is the curated answer fluent and coherent?
- (iii) Answer Comprehensiveness: Does the curated answer cover all aspects relevant to the question without missing pertinent information?
- (iv) Answer Conciseness: Does the answer only include the information necessary for the question without redundant content?
- (v) Question Relevance: Is the question relevant to real-life information-seeking encounters in the expert’s professional life?

We consider a QA pair to meet our quality requirement if it scores at least a 4 across all dimensions, following our experts’ recommendations.

3.4.2 Expert Evaluation Results

During iterations to improve our QA pair curation strategy, we had a legal expert perform open annotations on 50 curated QA pairs for each strategy. These 50 pairs were drawn from 10 different guides, with 5 QA pairs from each guide, using randomly selected consecutive paragraphs. An example of such an annotation can be found in Table A.10. When our expert annotated the resulting 50 QA pairs we derived the next strategy from the suggested improvements. The resulting scores, presented in Table 3.1, show the average performance of each strategy across all evaluation metrics. A detailed distribution of each metric is provided in Figure A.2.

Prompt	Question Relevance	Question Fluency	Answer Fluency	Answer Compr.	Answer Concise.
Single-Paragraph	4.10	4.92	4.80	3.70	3.92
Multiple-Paragraphs	4.70	4.96	4.80	4.00	3.60
Sentence-Level-CoT	4.92	4.94	4.62	3.82	4.46
SL-CoT with Search	4.86	4.92	4.56	4.26	4.50
Legal SL-CoT with Search v1	4.58	4.88	4.32	4.08	4.70
Legal SL-CoT with Search v2	4.84	4.84	4.68	4.40	4.66

Table 3.1.: Expert Evaluation Scores

The main observed drawback of the pairs curated with the *Single-Paragraph* strategy is their lack of comprehensiveness. This is expected as some paragraphs only contain parts of legal

reasoning processes. Nonetheless, this approach already resulted in numerous quality QA pairs. To increase the chances and allow the model to detect legal reasoning processes and curate more comprehensive answers for them, we developed our *Multiple-Paragraphs* strategy which allows the LLM to select paragraphs from a given set of 5 consecutive paragraphs. While we were able to observe slight improvements in comprehensiveness, we experienced a clear drop in conciseness. To tackle this drop in conciseness, we developed our *Sentence-Level-CoT* strategy in which the LLM can select individual sentences instead of entire paragraphs. A key observation by our legal expert during the evaluation of the first three strategies was that the paragraphs required for a fully comprehensive answer to the generated question were often distributed across different sections of the guide. Creating a general legal question that is precisely answered by the presented paragraphs is often not feasible, as some paragraphs only provide example cases or partial arguments related to broader legal issues. We, therefore, introduced the following three-step process: (i) generate a question based on randomly selected consecutive paragraphs, (ii) use semantic similarity search to identify potentially relevant passages, and (iii) curate an answer by generating it with sentence-level references from the guide, resulting in our *Sentence-Level-CoT with Search* strategy. With this, we were able to improve comprehensiveness while not reducing conciseness. The intermediary generation step (iii) was introduced to achieve a more natural flow in the answer and potentially better argumentation flow, as direct sentence selection in *Sentence-Level-CoT* consistently resulted in consecutive sentence picks (e.g., [1,3,7,9]), whereas a more natural flow could be achieved by selecting sentences out of order through answer generation with sentence-level citations (e.g., [1,9,7,3]). While we were not able to improve answer fluency, the intermediary generation step increased answer comprehensiveness resulting in better argumentation flow. We made a final effort to enhance the quality of the QA pairs by reformulating the prompts into a more formal legal terminology. We redesigned the question-generation prompts to produce questions that neither refer to specific cases nor ask for overly broad and abstract standards. Instead, the questions should connect fact patterns directly to pertinent legal doctrines, thereby establishing more effective general information-seeking scenarios. Overall, we did observe some improvements with our second version (*Legal Sentence-Level-CoT with Search v2*), while the first version (*Legal Sentence-Level-CoT with Search v1*) did not do any better than the non-legal *Sentence-Level-CoT with Search* in our evaluation.

3.4.3 Automatic Evaluation

Recent studies have proposed the use of LLMs to mimic human annotations and have shown good agreement rates (X. Li, Cao, et al., 2023; Pan et al., 2024; Zheng et al., 2024). To scale our expert evaluation and use it as a filtering strategy to increase the quality of our curated QA pairs in ECHR-QA, we are interested in whether GPT-3.5 can effectively evaluate the curated QA pairs across answer fluency, comprehensiveness, and conciseness. We used the 300 annotated QA pairs from our expert evaluation to create a filtering prompt based on G-EVAL (X. Li, Cao, et al., 2023). After testing both legal and non-legal zero-shot and few-shot CoT prompting strategies, we found that all performed similarly. However, using prompt A.11, a non-legal CoT prompt, we achieved the highest correlation with our expert

annotations. We report the F1-score, confusion matrix on the binarized value of low and high quality, and Kendall- τ correlation with p-value on the 300 annotated QA pairs in Table 3.2. The results show that we were not able to replicate our expert annotations well, however, our automated assessment helps us identify and filter some low-quality samples, increasing the overall percentage of high-quality pairs from 55% to 58%. If we only consider the 150 QA pairs from the search-based prompts, 67% are deemed of high quality before filtering, which, on filtering, raises to 72%. Although we anticipated that comprehensiveness would be the most challenging metric for an LLM to evaluate, it turned out to be the only metric where our filtering approach shows moderate correlation.

Category	F1 Score	TP	FP	TN	FN	Kendall's Tau	p-value
Fluency	0.96	281	14	1	4	0.04	0.40
Comprehensiveness	0.82	195	75	23	7	0.27	$8e^{-08}$
Conciseness	0.78	179	42	18	61	0.05	0.37
Overall	0.63	116	83	50	51	0.169	0.0008

Table 3.2.: Filtering Results

3.5 ECHR-QA Dataset

To generate our final dataset, we iterated over all guides using a context window of three paragraphs with a stride of two paragraphs. We then attempted to create QA pairs that passed our filtering criteria using our three search-based prompts (see Code A.12). First, we used *Legal Sentence-Level-CoT with Search v2* to generate a QA pair, followed by our filter to assess its quality. If the pair did not achieve 4 or higher across answer fluency, comprehensiveness, and conciseness, we repeated the process with *Legal Sentence-Level-CoT with Search v1*, and finally with *Sentence-Level-CoT with Search*. If none of the generated QA pairs passed the filter, we moved on to the next set of three paragraphs. We imposed additional requirements during filtering to further improve the quality and relevance of the dataset. First, we filtered out duplicate questions by calculating cosine similarity between each question and all previous ones, using `text-embedding-3-small` embeddings with a threshold of 0.8. This step was implemented to ensure that only unique questions were included in the dataset. Additionally, we excluded QA pairs where the citations in the answers referenced cases that did not have an available English translation. This was necessary because the hyperlinks in the case-law guides could lead to judgments in either English or French, the two official languages of the Court. Furthermore, we filtered out pairs where the citations did not include pinpointed paragraph numbers but rather cited an entire case. This criterion was important to maintain the specificity and precision of the references within the dataset. To ensure the questions in our dataset were general and did not refer to specific cases, we filtered out any questions that mentioned cases, to ensure questions focus on general legal principles and guidelines rather than case-specific details.

Our final dataset⁵ comprises 1116 QA pairs. On average, each question contains 49 tokens⁶, each answer contains 295 tokens and 4.9 sentences, and each answer cites an average of 7.8 paragraphs. The distribution of these statistics is illustrated in [A.1](#).

To gain a deeper legal understanding of the dataset, as well as insights into some of its benefits and drawbacks, we provide a legal case study of QA pairs from the dataset, with examples in our Appendix under [A.1 ECHR-QA Dataset Case Study](#).

⁵https://github.com/KorbiQWeidinger/echr-qa/blob/master/data/echr_qa_dataset.csv

⁶LLMs have a predefined vocabulary of tokens; we obtained tokens with <https://www.npmjs.com/package/tiktoken>

4 Attributed QA Approaches

This section describes all approaches from which we sampled and evaluated responses for the ECHR-QA dataset. We conducted 16 total experiments with 4 different models, 4 different approaches, and 1 retriever. All experiments produce an answer to a given question and attributions in the form of in-line citations (see [3.1 ECHR-QA Task](#)). The benchmark results of our experiments can be found in Chapter 5. The code to reproduce our experiments can be found on GitHub¹.

4.1 Retrieval

We use the SQLite Database v2.0.0 by ECHR Opendata² ([Quemy et al., 2021](#)) offering 16,096 English cases. To have the possibility to produce answers with citations close to the answers in ECHR-QA, we made sure our approaches had access to all citations from the ECHR-QA dataset. We therefore added 162 cases from the web that were missing in the ECHR open data database but were in our ECHR-QA dataset. Our final database (DB) consists of 16,258 cases with slightly over 1 million paragraphs. We use GTR ([Ni et al., 2022](#)) with `gtr-t5-xl`³ to retrieve paragraphs.

4.2 Models

We aim to evaluate the performance of 4 Open Source LLMs on ECHR-QA. We chose Mistral-7B⁴ ([Jiang et al., 2023](#)) and SaulLM-7B⁵ ([Colombo et al., 2024](#)), alongside Llama-3-8B⁶ and Llama-3-70B⁷ ([Touvron et al., 2023](#)). Each model was selected based on unique characteristics which we want to investigate.

SaulLM-7B is an extension of Mistral-7B, further specialized by additional unsupervised training on an English legal corpus followed by instruction tuning. SaulLM-7B has outperformed other models on LegalBench-Instruct. Hence, we expect it to outperform Mistral-7B and other models

¹<https://github.com/KorbiQWeidinger/echr-qa>

²<https://echr-opendata.eu/>

³<https://huggingface.co/sentence-transformers/gtr-t5-xl>

⁴<https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.2>

⁵<https://huggingface.co/Equall/Saul-7B-Instruct-v1>

⁶<https://huggingface.co/meta-llama/Meta-Llama-3-8B>

⁷<https://huggingface.co/meta-llama/Meta-Llama-3-70B>

given that ECHR-QA is a legal task. On the other hand, Llama-3-8B and Llama-3-70B offer a comparative view of how model size impacts performance.

4.3 Approaches

RAG (Lewis et al., 2020): Our simple RAG-based approach retrieves $k = 10$ paragraphs from the DB based on the given question. The LLM is then prompted with the question and the retrieved passages to produce the answer (Prompt A.14) and tasked to accompany sentences with inline citations to these paragraphs.

LLattribution (X. Li, Zhu, et al., 2023): The retrieval process from RAG may fail to retrieve the evidence required to provide a comprehensive and attributable answer, resulting in a response that is not as optimal as it could be. This, in turn, may undermine the abilities of the LLM. To address this shortcoming, X. Li, Zhu, et al., 2023 propose utilizing the LLM to update the retrieval result until it verifies that the retrieved paragraphs can sufficiently support answering the question.

We follow a slight modification of the missing-info-query approach proposed by X. Li, Zhu, et al., 2023 as we additionally do an initial retrieval ($k = 8$) based on the question. Following the initial retrieval, the process proceeds with the iterative loop proposed by X. Li, Zhu, et al., 2023. This loop comprises three key steps:

- (i) Retrieval based on a missing-information query (Prompt A.15)
- (ii) Sliding document selection (Prompt A.16)
- (iii) Evaluation of the selected paragraphs (Prompt A.17)

In step (i), the LLM is prompted to identify and provide information missing in the current set of retrieved paragraphs. We then retrieve $k = 10$ paragraphs based on the provided missing information. While we have more than 10 total paragraphs, a sliding window approach (ii) is employed to refine the selection to a maximum of 10. Once the LLM has narrowed the selection to 10 or fewer paragraphs, it is tasked with evaluating their sufficiency to answer the query. This evaluation involves assigning a score between 1 and 10, with a minimum score of 8 required to exit the retrieval loop. If this threshold is not met, the loop continues for up to three iterations.

The final answer is generated using the prompt provided in A.14, which is the same prompt utilized in RAG.

Post-hoc: In this approach, the LLM is first used to generate an answer given the question (Prompt A.13), without any use of retrieval. We generally refer to this as the base or closed-book answer. We then use GTR to retrieve the most relevant passage for each sentence in the generated answer, citing it only if the similarity score is 0.5 or greater.

RARR (L. Gao et al., 2022): Since LLMs often generate unsupported or misleading content, L. Gao et al., 2022 proposed a research-and-revise strategy where the evidence retrieved

based on the initial response is reviewed, and the response is edited to correct unsupported content and make it consistent with the evidence. For simplicity, we used a zero-shot strategy as opposed to the view-shot strategy proposed by [L. Gao et al., 2022](#). In our setting, a view-shot strategy would result in a much larger token window and would require consultation with our legal expert to write the view-shot prompts. For each sentence in the generated response, we first prompt the LLM if supporting evidence is required for the sentence (Prompt: [A.18](#)). If supporting evidence is required, we retrieve $k = 3$ passages and use the LLM to identify which support, contradict, or are irrelevant to the sentence (Prompt: [A.19](#)). If we find supporting passages, we include the sentence with them as citations; if we find no supporting but only contradicting passages, we edit the sentence (Prompt: [A.20](#)).

5 Evaluation

This chapter describes the utilization of ECHR-QA as a benchmark for LLMs. It introduces a framework for automatic evaluation designed to assess responses to questions posed in ECHR-QA, spanning four key dimensions: answer fluency, correctness, groundedness, and citation relevance. Subsequent sections discuss the findings from the automatic evaluation conducted on the experiments detailed in Chapter 4. We then present a comparative analysis between expert annotations and the automatic evaluation method regarding answer correctness, groundedness, and citation relevance.

5.1 Automatic Evaluation

Task Setup

We extend the task of attributable legal question answering for evaluation as follows: Given a legal query q , a target answer TA extracted directly from the case law guides, its citations TC with their texts from ECHR judgments, and the generated answer GA with its citations GC obtained from ECHR judgments (see 4.1 Retrieval), we aim to automatically measure the quality of the generated answer and its citations. Each answer consists of sentences $TA = \{ta_1, \dots, ta_n\}$ and $GA = \{ga_1, \dots, ga_k\}$. Sentences in a GA have citations $GC_i = \{gc_{i,1}, \dots, gc_{i,j}\}$ where each citation gc is a triple of case id, paragraph number, and the paragraph itself $gc_{i,j} = (gc_{i,j}[id], gc_{i,j}[pn], gc_{i,j}[par])$. Each TA comes with citations TC at the answer level where each tc_i is a triple of case id, paragraph numbers, and paragraphs $tc_i = (tc_i[id], tc_i[pns], tc_i[pars])$.

Metrics

To automatically evaluate the quality of the generated responses, we propose the following metrics for comparing them to the target response. These metrics aim to capture answer fluency and correctness, as well as citation groundedness and relevance.

Answer Fluency: We employ MAUVE (Pillutla et al., 2021), a metric designed for evaluating open-ended text generation to assess the fluency of generated answers. MAUVE measures how closely the distribution of text generated by a model aligns with the distribution of human-written text using divergence frontiers. To calculate the Mauve score for a GA , we follow T. Gao et al., 2023 and compare the TA from ECHR-QA, whose sentences are

human-written, with the GA and truncate both at 100 words. As LLMs are capable of producing fluent text, MAUVE primarily functions as a sanity check, given its sensitivity to output length and text style (T. Gao et al., 2023).

Answer Correctness: The goal of measuring correctness is to approximate the informativeness and utility of the generation to the question as perceived utility by humans, a process difficult to automate (T. Gao et al., 2023; Krishna et al., 2021). Previous work has measured the correctness of generated answers by comparing them to ground truth answers (Funkquist et al., 2022; T. Gao et al., 2023). In similar contexts, ROUGE measures (Lin, 2004), BERTScore (T. Zhang et al., 2019), and NLI models have been proposed to assess answer correctness (Funkquist et al., 2022; T. Gao et al., 2023). We provide ROUGE-L¹ as it is effective for capturing surface-level similarities between generated texts and references, such as the presence of specific key phrases or n-grams; it primarily focuses on the lexical overlap and doesn't account for semantic accuracy or the contextual appropriateness of the text (Krishna et al., 2021; Lin, 2004). Therefore, ROUGE metrics struggle to represent correctness well (Krishna et al., 2021; T. Zhang et al., 2019). We also provide the BERTScore² F1 values, a more recent alternative to ROUGE that uses contextual embeddings to calculate a similarity score between the GA and the TA. BERTScore addresses some limitations of ROUGE by using contextual embeddings to assess the semantic similarity between generated and reference texts (Funkquist et al., 2022). However, BERTScore still struggles with deeper linguistic nuances and contextual relevance (Hanna et al., 2021). We also provide the BERTScore using T5-large (Raffel et al., 2020) as the embedding model. The NLI approach $\phi(\text{premise}, \text{hypothesis}) \in \{0, 1\}$ proposed in similar work, in particular, with models of the TRUE family, to measure factual correctness addresses this shortcoming (Bohnet et al., 2022; T. Gao et al., 2023). We use TrueTeacher³ (Gekhman et al., 2023) to calculate the *claim recall* (T. Gao et al., 2023) of generated answers, to assess whether each claim in the target answer (hypothesis) is entailed by the entire generated answer (premise) and report the average entailment score across all claims in each target answer.

$$\text{claim recall} = \frac{\sum_{ta_i \in TA} \phi(GA, ta_i)}{|TA|}$$

Groundedness: We additionally aim to evaluate each response's groundedness. A response is considered grounded if the key propositions in its response have valid references to relevant legal documents (Magesh et al., 2024). We follow T. Gao et al., 2023 and calculate citation faithfulness recall and precision to evaluate attributions. *Citation faithfulness recall* describes the average entailment of a sentence to its citations. *Citation faithfulness precision* is averaged over all $gc_{i,j}$, where the precision of $gc_{i,j}$ is 1 if it is not "irrelevant" and recall is 1. A citation is "irrelevant" if it does not support the statement $\phi(gc_{i,j}, ga_i) = 0$ and it is not needed to support the statement $\phi(GC_i \setminus gc_{i,j}, ga_i) = 1$. *Citation faithfulness precision* detects irrelevant citations

¹<https://pypi.org/project/rouge-score/>

²https://github.com/Tiiiger/bert_score

³https://huggingface.co/google/t5_11b_trueteacher_and_anli

but does not require citing a minimal set (T. Gao et al., 2023). We again use TrueTeacher (Gekhman et al., 2023) for all NLI measures.

$$\begin{aligned} \text{citation faith. recall} &= \frac{\sum_{ga_i \in GA} \phi(GC_i, ga_i)}{|GA|} \\ \text{irrelevant}_{i,j} &= (\phi(gc_{i,j}, ga_i) = 0) \wedge (\phi(GC_i \setminus gc_{i,j}, ga_i) = 1) \\ \text{citation faith. precision} &= \frac{\sum_{i \in GA} \phi(GC_i, ga_i) \cdot \sum_{j \in GC_i} (1 - \text{irrelevant}_{i,j})}{\sum_{i \in GA} |GC_i|} \end{aligned}$$

Citation Relevance: To approximate the relevance of the citations used in the generated answers, we measure the exact match (EM) of citations in the generated response compared to the target response, and report the F1-score, following (X. Li, Cao, et al., 2023). Since multiple cases can contain the same information and there is a selective citation bias in the case-law guides (Santosh, Haddad, et al., 2024; Santosh, Hernandez, et al., 2024), we also compute citation entailment using NLI-based TrueTeacher (Gekhman et al., 2023) between the citation paragraphs in the generated and the target response. If there are multiple cited paragraphs from the same case, we concatenate these paragraphs to treat them as a single text block tcb of citations and assess whether each reference citation block is covered by the generated citations. Specifically, for each target citation block, we check if there exists a generated citation that is entailed by the target citation block and report the average recall across all target citations.

$$\text{citation relevance recall} = \frac{\sum_{tcb \in TC} \max_{gc \in GC} \phi(tcb, gc)}{|TC|}$$

5.2 Results

Answer Fluency

All models achieve good fluency scores. In the closed-book strategies Post-hoc and RARR, SaulLM-7B consistently achieves higher MAUVE scores compared to other models. This capability is likely attributed to its specialized legal pre-training, which enables the SaulLM-7B model to generate text that closely mirrors the formal and technical language characteristic of legal discourse. In contrast, within RAG and LLattribution, all other models show higher scores than SaulLM-7B, demonstrating their adaptability to adjust to the style dictated by the retrieved context, with the Llama family generally outperforming Mistral in most cases. As anticipated, the retrieve-then-generate approaches yield higher MAUVE scores, as they can adapt to the writing style of the retrieved ECHR cases, which closely align with the formulations found in case-law guides.

Strategy	Model	Fluency	Answer Correctness			Citation Faithfulness		Citation Relevance	
		Mauve	R-L	BERT S	NLI	Prec.	Rec.	EM	NLI
RAG	Mistral-7B	89.6	23.7	58.0	11.5	61.4	54.2	6.0	46.0
	SaulLM-7B	76.7	21.2	55.2	10.0	17.8	14.6	3.6	28.5
	Llama-3-8B	91.7	25.9	58.4	9.6	70.3	67.3	6.5	43.8
	Llama-3-70B	88.1	26.4	58.6	9.4	72.2	68.5	7.3	45.7
LLatriveal	Mistral-7B	79.8	24.7	58.4	11.2	76.7	69.7	5.7	49.0
	SaulLM-7B	73.0	21.0	54.4	7.9	19.2	16.9	2.4	27.2
	Llama-3-8B	86.3	25.9	58.4	9.7	74.2	72.4	5.6	39.7
	Llama-3-70B	88.4	26.7	58.9	9.4	73.2	70.3	7.8	46.7
Post-hoc	Mistral-7B	71.2	19.2	55.6	5.5	4.4	4.4	0.3	25.4
	SaulLM-7B	78.3	21.1	55.0	4.3	4.3	4.3	0.3	19.3
	Llama-3-8B	63.0	18.9	54.7	4.3	4.0	4.0	0.3	32.3
	Llama-3-70B	66.5	19.4	55.3	5.8	4.1	4.1	0.2	24.4
RARR	Mistral-7B	71.2	19.2	55.6	5.5	25.5	21.5	3.0	70.7
	SaulLM-7B	78.3	21.1	55.0	4.3	19.2	16.6	3.1	58.9
	Llama-3-8B	63.1	18.9	54.7	3.5	23.6	21.2	3.0	59.0
	Llama-3-70B	66.5	19.4	55.3	5.8	31.7	25.6	3.1	66.6

Table 5.1.: Comparison of different LLMs across attributed QA strategies on ECHR-QA

Answer Correctness

Our closed-book approaches, Post-hoc and RARR, exhibit lower performance scores compared to our open-book methods, RAG and LLatriveal, across all three answer correctness metrics. This suggests that the retrieve-then-generate strategies produce answers that align more closely with the target answers in ECHR-QA. Such alignment is expected, given that the target answers in ECHR-QA are derived directly from the case-law guides, which closely resemble the content of ECHR judgments.

In terms of ROUGE-L scores, SaulLM outperforms other models within the closed-book approaches, while performance metrics such as BERTScore and our NLI-based claim recall see comparable or better results in other models. This indicates that SaulLM’s legal pre-training aids in generating legal terms but may lack other essential aspects required for answering the questions like superior legal reasoning. SaulLM notably has a lower claim recall than Mistral, further indicating that its legal pre-training has not improved its legal question-answering capabilities. Model size however does seem to increase legal question-answering capabilities, with the 70B Llama-3 model showing higher scores than the smaller 7B model across ROUGE-L, BERTScore, and claim recall in both Post-hoc and RARR.

Our RARR implementation does not seem to be able to improve answers, and generally only leads to very few edits (see Table 5.2). This implies that our implementation of RARR is not

able to utilize the retrieved cases to improve the answer. The 3% of edited answers in RARR Llama-3-8B even negatively impact answer correctness. As shown in our case study A.2, the LLM often just rewrites sentences without changing the meaning. We can see that RARR has the potential to improve answers, but it is not able to do so consistently. In some cases, it even makes wrong edits as shown in Table A.15.

In our open-book strategies, Llama-3-70B achieves the highest ROUGE-L and BERT-Scores, while Mistral achieves the best claim recall. Again the legal pre-training seems to not have helped SaulLM in the open-book strategies, and it consistently performs worse than Mistral.

Model	Edited Answers	Total Edits	Total Sentences
Mistral-7B	0.1%	18	17479
Saul-7B	0.5%	55	11559
Llama-3-8B	3%	727	22447
Llama-3-70B	0.09%	22	23880

Table 5.2.: Edits by RARR

Groundedness

The open-book strategies yield much higher citation faithfulness scores compared to the closed-book strategies. Simply using GTR to find citations (Post-hoc) works very poorly, as the low numbers indicate. Even RARR, which allows the LLM to decide if a retrieved case supports a claim, does badly compared to the open-book strategies.

LLatriveal seems to be able to increase claim recall and precision over RAG, indicating a decrease in hallucinations where claims have citations but the source does not support the claim. However, the average citations per answer decrease drastically while answer length only decreases slightly with LLatriveal (see Table 5.3), which could imply that the number of ungrounded claims increases - a measure neither captured by claim recall nor precision.

SaulLM shows surprisingly low values for citation faithfulness; an investigation of its responses showed that it did not follow our instructions well compared to the other models, hallucinating cases that were not retrieved, or not adding the citations to the corresponding sentences as instructed. This impacts all approaches except Post-hoc SaulLM, as all strategies are very instruction-driven and rely on models following the presented output format. We indicated the impacted SaulLM results in Table 5.1 with red.

RAG and RARR values for Llama 7B and 70B indicate that model size increases citation faithfulness, but surprisingly we do not observe the same for LLatriveal.

Citation Relevance

We can observe that RAG and LLatriveal seem to produce similarly relevant citations. Notably, RAG has a slightly higher average number of citations, increasing the chance that our NLI-based *citation recall* measure finds a gc for a target citation block tcb . This indicates that

Strategy	Model	Tokens	Sentences	Citations
RAG	Mistral-7B	265	6.1	6.0
	SaulLM-7B	332	6.9	3.4
	Llama-3-8B	238	4.9	4.8
	Llama-3-70B	179	3.8	5.7
LLat retrieval	Mistral-7B	252	5.7	4.7
	SaulLM-7B	314	6.5	2.6
	Llama-3-8B	224	4.8	3.8
	Llama-3-70B	175	3.7	4.7
Post-hoc	Mistral-7B	479	15.7	3.4
	SaulLM-7B	298	10.4	2.4
	Llama-3-8B	506	20.1	4.3
	Llama-3-70B	514	21.4	3.5
RARR	Mistral-7B	479	15.7	18.0
	SaulLM-7B	298	10.4	12.7
	Llama-3-8B	506	20.1	11.9
	Llama-3-70B	514	21.4	15.5

Table 5.3.: Comparison of avg. answer tokens, sentences, and citations of different LLMs across attributed QA strategies

generated citations in LLat retrieval might be slightly more relevant compared to RAG. This is further supported by little to no drops in EM values between LLat retrieval and RAG. Mistral achieves the best *citation recall* measures while Llama 70B achieves the best EM results for RAG and LLat retrieval. Notably, there seems to be no overall correlation between our EM and NLI-based metrics for citation relevance. SaulLM again shows comparably low values, likely again due to not adding the citations to the corresponding sentences as instructed. While post-hoc performs very poorly in both EM and NLI, RARR achieves the highest NLI values. However, RARR also includes on average almost 2-3 times as many citations compared to RAG and LLat retrieval, increasing its chances to find a *gc* for each *tcb* drastically. Since our citation measure is a recall measure, RARR is not punished for including irrelevant citations during citation relevance evaluation. This observation is, however, further supported by RARR’s lower citation faithfulness values.

5.3 Expert Evaluation

We conducted blind expert annotations for a total of 40 randomly sampled question-answer pairs from ECHR-QA⁴. In each annotation, the expert is tasked to rank 4 generated answers

⁴We do require the 4 experiments to have a minimum of 2 different claim recall values to be able to get an automatic ranking for them. It is important to mention that this is only true for a minority of QA pairs as 73% of generated answers across all experiments have a claim recall of 0.

(A, B, C, and D) across three dimensions. These dimensions correspond with the dimensions from our automatic evaluation framework. The answers are shuffled to make it harder for the expert to identify which experiment resulted in a presented answer. We have developed a custom interface for these annotations, which is showcased in Figures A.21 and A.22. We allow for equal rankings, as two answers might be equally correct, have the same amount of hallucinations, or cite the same sources. The dimensions as depicted in the annotation platform are:

- (i) Correctness: Rank the experiments based on the accuracy, completeness, and relevance of their answers to the given question.
- (ii) Groundedness: Rank the experiments by how constitutively their answers are supported by citations.
- (iii) Source Relevance: Rank the experiments by how relevant and directly supportive the cited sources are in answering the question.

Additionally, the expert may submit a binary value to indicate whether a generated answer surpasses the target responses from the ECHR-QA dataset (Table 5.7). Lastly, the interface supports the incorporation of detailed notes to all presented question-answer pairs. The experts' thoughts and findings are shared in A.2 Attributed QA Case Study. The code for the platform and the experts' rankings are publicly available on GitHub⁵.

Strategy	Correctness		Groundedness		Citation Relevance	
	Expert	Auto	Expert	Auto	Expert	Auto
RAG	2.45	2.10	2.15	1.70	2.15	2.67
LLattribution	1.80	2.00	2.10	1.43	2.15	2.20
Post-hoc	2.90	2.95	3.30	3.90	3.35	3.15
RARR	2.85	2.95	2.45	2.98	2.35	1.98

Table 5.4.: Average expert ranking compared to auto ranking for Mistral-7B across methods

Model	Correctness		Groundedness		Citation Relevance	
	Expert	Auto	Expert	Auto	Expert	Auto
Mistral-7B	2.20	2.27	2.45	2.65	2.45	2.27
SaulLM-7B	2.75	2.25	2.55	3.42	2.55	2.75
Llama-3-8B	2.30	2.73	2.65	2.15	2.65	2.40
Llama-3-70B	2.75	2.75	2.35	1.77	2.35	2.58

Table 5.5.: Average expert ranking compared to auto ranking for RAG across models

We first presented our expert 20 QA pairs with experiment results from Mistral across strategies 5.4. Average automatic rankings are calculated from claim recall, citation faithfulness recall, and

⁵<https://github.com/KorbiQWeidinger/echr-qa-annotation-platform>

citation relevance recall for Tables 5.4 and 5.5. Across strategies (Table 5.4), RAG paradigms perform well, with LLattribution taking the lead with the best correctness and groundedness according to our expert. This aligns with the observable trend from our automatic evaluation, where RAG approaches consistently outperform post-hoc methods. Notably, our expert also prefers RAG approaches over RARR for citation relevance. As previously discussed, our methodology of calculating citation relevance recall exclusively, which searches for generated citations entailed within target citation blocks, is benefiting RARR in this context. Therefore, we recommend calculating citation relevance precision values in future work to provide a more comprehensive evaluation of the effectiveness of various citation generation methodologies. Table 5.5 presents 20 more expert annotations for vanilla RAG across models. The table, in conjunction with our automatic evaluation, demonstrates that comparing models within the same strategy proves to be more challenging than comparing across different strategies, as the performance metrics are significantly closer. Generally, Mistral achieves the highest claim recall in RAG, as shown in Table 5.1. This trend continues with our expert slightly favoring Mistral as the model with the best answer correctness within RAG. Both automatic evaluation as well as our expert favor Llama-3-70B for groundedness. For citation relevance, our automatic evaluation favors Mistral, while our expert slightly favors Llama 70B again.

Evaluation	Metric	Corr.	p-value
Answer Correctness	ROUGE-L	0.09	0.23
	BERTScore	0.23	0.0028
	Claim Recall	0.09	0.24
Groundedness	Faith. Prec.	0.32	$3.8e^{-5}$
	Faith. Rec.	0.26	0.0008
Citation Relevance	EM	0.10	0.19
	Recall	0.38	$5.38e^{-7}$

Table 5.6.: Correlation and p-value of expert rankings compared to auto rankings

We calculate the Spearman rank correlation coefficient (Spearman, 1961) across all rankings derived from each of the 40 annotations. Correlations for all metrics are documented in Table 5.6. For answer correctness, claim recall surprisingly shows no correlation. It is important to note, that 73% of generated answers have a claim recall of 0. This suggests that calculating claim recall at the sentence level is not very effective. Sentences can encompass multiple claims and examples, potentially reducing the meaningfulness of sentence-based claim recall values compared to calculations performed at the individual claim level, the approach suggested by T. Gao et al., 2023. Surprisingly, BERTScore demonstrates better correlations, potentially lending further support to this claim, whereas ROUGE-L also exhibits no correlation. For groundedness, both citation faithfulness precision and recall demonstrate moderate correlations, as does our NLI-based citation relevance recall. In contrast, our exact match (EM) metric for measuring citation relevance shows no correlation, further highlighting the potential advantages of using an NLI-based approach.

As previously mentioned, we asked our legal expert to assess which of the 160 generated

Strategy	Model	Better than Target Answer
RAG	Mistral-7B	15.00%
	Saul-7B	30.00%
	Llama-3-8B	20.00%
	Llama-3-70B	15.00%
RARR	Mistral-7B	15.00%
Post-hoc	Mistral-7B	15.00%
LLatriveal	Mistral-7B	25.00%
Total		18.75%

Table 5.7.: Percentage of answers the expert labeled as better than the target answer from the 160 generated answers in the 40 annotations.

answers presented during the evaluation were generally better than the ECHR-QA target answers curated during dataset generation (Chapter 3). Our expert found 30 of the 160 presented generated answers (18.75%) to be better than the target answers. Our expert found 6 of the 20 answers generated by Saul to be superior to the corresponding target answers, suggesting that the ability to produce better-than-target answers does not strongly correlate with the model’s overall ability to consistently generate high-quality responses. Mistral was generally preferred over Saul, indicating that this outcome may be more influenced by luck and the quality of the target answers than by the model’s inherent capabilities. In 2 of the 40 presented ECHR-QA tasks, all generated answers were preferred over the target answer. This is somewhat expected, as approximately 30% of the curated QA pairs in ECHR-QA are anticipated to be of medium to low quality, as highlighted in Section 3.4.3 of Chapter 3. To gain a deeper understanding of the dataset and the generated responses, we present a second case study conducted by our ECHR expert (Appendix A.2).

6 Conclusion

We curated ECHR-QA, a benchmark dataset for attributed legal long-form question answering focused on ECHR jurisprudence, with generated questions and extracted answers from case-law guides, using large language models (LLMs) and expert quality evaluation. We explored various QA pair curation strategies, including single-paragraph, multiple-paragraphs, and sentence-level approaches, ultimately refining our method to a legal sentence-level chain-of-thought with search strategy that improved question quality and answer comprehensiveness. Additionally, we made a moderately successful attempt to leverage LLMs as reference-free evaluators to filter out low-quality data.

Our legal expert’s assessment revealed that the dataset is predominantly of high quality, although a notable portion of the data may not fully meet the standard of providing perfectly comprehensive answers to the generated legal queries. The legal expert’s analysis highlights significant limitations in the LLMs’ comprehension of legal language and the nuances of legal judgments.

We then used four different open-source LLMs, including a legally trained one, across four distinct strategies for attributed long-form question answering: RAG, LLattribution, Post-hoc attribution, and RARR to answer the generated questions from ECHR-QA.

We evaluated the performance of the LLMs and strategies using an automatic evaluation approach. Our evaluation revealed several key insights and challenges in the field of attributed legal question answering. Firstly, we observed that models with domain-specific legal training, while potentially beneficial, can sometimes overfit to their training data, limiting their generalization capabilities. Secondly, the retrieval of relevant evidence based on semantic similarity remains a significant challenge, highlighting the complexity of legal language and the need for more sophisticated retrieval mechanisms. Lastly, our evaluation underscored the difficulties that LLMs face in accurately identifying and leveraging supporting evidence, a crucial aspect of providing reliable and attributable legal answers.

Our legal expert generally found RAG-based answers to be impressive, noting that they usually provided the sought-after fundamentals of the questions while often lacking specific details and comprehensive examples.

We hope that our benchmark, ECHR-QA, will benefit the legal NLP community by fostering the development of improved methods for long-form question answering, providing both faithful responses and verifiable attribution.

7 Limitations

In this chapter, we highlight some of the known limitations and drawbacks of the ECHR-QA dataset, our experiments, and the automatic evaluation approach.

ECHR-QA Dataset

The first obvious limitation is that the generated questions can contain mistakes or lack relevance, both of which were observed by our expert. A clear limitation of the sentence-based extraction approach used in our final dataset generation is that the order of sentences in extracted answers can lead to errors; however, this was rarely observed by our expert. Both the question generation and our filtering attempts revealed significant limitations in GPT-3.5’s comprehension of legal language and the nuances of legal judgments, which made the generation of a high-quality dataset challenging. In some cases, however, the expert also attributed the model’s inability to construct accurate responses to poor drafting in the case-law guides themselves. Based on the evaluation of our prompts, the dataset is expected to have only around 70% of QA pairs meeting our expert’s required quality standards.

Evaluation of QA Pair Curation Strategies

A potential threat to the validity of the improvements presented in Table 3.1 is that the annotations were conducted iteratively while refining the strategies and prompts. This introduced the possibility of bias, especially since the expert involved in the annotations also contributed to writing the prompts.

Attributions in ECHR-QA

There are multiple limitations in our citation identification approach in ECHR-QA. In the context of case-law guides, citations are typically provided as hyperlinks leading directly to the corresponding case in the HUDOC database. However, it is important to note that not all these links point to judgments available in English. When encountering a citation pointing to a case in French, we attempt to find its English counterpart. If this is not feasible, the curated QA pair will be excluded from the ECHR-QA dataset, as we require all cases within ECHR-QA to be available in English.

There are scenarios where multiple consecutive sentences belong to the same reference. In such cases, it is often acceptable to place the citation at the end of the sequence of sentences,

rather than after each sentence. While this practice is straightforward for human readers to interpret, it poses significant challenges for automated systems. Consequently, our approach assumes that a citation is directly linked to the sentence it is included in, but not to other sentences that may precede or follow. This implies that a sentence indirectly related to a cited source may appear without a citation in the dataset.

It is crucial to recognize this approach's limitations, particularly in automated text processing and the potential for incomplete attribution in sequences of related sentences. This nuanced handling of citations underscores the complexities inherent in developing comprehensive and accurate case-law datasets. In the future development of attributed legal QA datasets based on case-law citations, one could attempt to leverage Natural Language Inference (NLI) models to enhance the accuracy of citation attributions.

Despite all these challenges, we were able to generate a dataset of largely good quality. We will discuss potential improvement suggestions in Chapter 9.

Limitations of LLMs

The legal expert's analysis highlights significant limitations in the LLMs' comprehension of legal language and the nuances of legal judgments. The used LLMs struggle to discern legal norms and the factors that shape them, often failing to adapt arguments based on subtle details. Our expert found them unable to effectively connect general descriptions of legal articles or general doctrinal statements with specific circumstances that inform their application, thereby missing the crucial relationship between facts and legal precedent that make up legal decision-making. Moreover, the models demonstrate errors such as the inability to extract concepts linking precedents, leading to challenges in understanding how different aspects of an Article relate to each other or apply to other Articles. They also struggle to differentiate between various legal components such as situations, standards, criteria, conditions, reasons, and restrictions. Consequently, their responses either lack necessary details about the situation presented in the question or provide overly generalized answers that overlook the specific context, rendering them ineffective for practical legal analysis.

In essence, these limitations underscore the need for further development in LLMs to better grasp the complexities of legal language and reasoning, enabling them to produce more nuanced and contextually relevant responses in legal applications.

Limitations of the Automatic Evaluation Approach

The first limitation of our automatic evaluation is the quality of the QA pairs in ECHR-QA. Claim recall is highly dependent on the inclusion of all relevant claims in the target answer. As previously discussed, some answers may not encompass all relevant claims. Another limitation of claim recall is its calculation at the sentence level, which can present challenges such as multiple claims being embedded within a single sentence and a lack of differentiation between claims and examples. Consequently, a long sentence containing multiple claims and an example might not be recognized as fully included, even though each claim individually might be, and a comparable example may also be present. However, the decision not to

extract claims for our dataset was intentional, as doing so could introduce an additional layer of complexity and potential errors.

The quality of attributions, on the other hand, primarily affects our citation relevance metric. As discussed previously, we did not consider indirect citations and encountered other limitations during citation identification. Thus, it is likely that some QA pairs do not include the actual full citation set. However, we believe that this limitation only slightly affects the calculation of citation relevance.

Citation faithfulness metrics are not impacted by dataset quality; however, we faced other limitations. Not enforcing structured LLM outputs¹ means that output formats can differ from the expected ones, leading to the incorrect association of attributions with sentences. We explicitly ask models to put citations at the end of a sentence; however, especially Saul struggled with this, resulting in low citation faithfulness and relevance scores. Often models would output something like *Used Documents: [1,2,3]* at inference, leading to a meaningless NLI measurement of $\phi("...", "Used Documents : ")$. We did make some attempts to remove repeated mistakes; however, without enforcing output formats, malformed output is still likely to occur. Enforcing output standards, however, might no longer accurately reflect true model performance.

Another limitation of our attempt to measure groundedness is the inability of our framework to capture ungrounded claims — key factual propositions that are not cited (Magesh et al., 2024). This dimension would provide insights into the proportion of claims in responses that lack supporting citations. Furthermore, our evaluation metrics heavily rely on NLI models for assessing different dimensions, where each part of the claim must be verifiable with evidence. However, human judgments often incorporate implicit world knowledge and may not require substantiating evidence for every aspect of a claim. Additionally, our evaluation approach lacks the capability to measure legal structures and reasoning, a very important aspect of legal question answering. Future research should explore methods to address these limitations and ensure a more holistic evaluation approach.

Limitations of Expert Evaluations

A significant limitation of our study is the reliance on a single expert for evaluations and annotations. While this approach provides valuable insights, it introduces several potential issues that warrant consideration. As previously mentioned, the annotations were conducted iteratively while refining the strategies and prompts, which introduced the possibility of bias, especially since the expert involved in the annotations also contributed to writing the prompts. Beyond this specific concern, the use of a single expert presents broader limitations. Regardless of their expertise, one individual may not represent the full spectrum of legal interpretations and perspectives. Different experts might have varying opinions on the quality and relevance of QA pairs, potentially leading to different evaluations.

Furthermore, the lack of multiple experts hampers our ability to assess inter-annotator agreement or cross-validate the evaluations. This limitation might affect the reliability of our expert

¹ Structured outputs can be achieved with paradigms like <https://github.com/guidance-ai/guidance>

evaluation results, making it more challenging to draw broad conclusions with high confidence. Another consideration is the workload placed on a single expert. Evaluating a large dataset can lead to fatigue over time, potentially affecting the consistency and quality of annotations. This is particularly relevant for extensive datasets like ours, where maintaining consistent evaluation criteria throughout the process can be challenging.

Lastly, while our expert is highly knowledgeable in ECHR case-law, they may have varying levels of expertise across different areas within this domain. This variation could influence the evaluation of certain QA pairs, potentially leading to inconsistencies in the assessment of more specialized or niche legal questions.

These limitations suggest that future work should ideally involve multiple experts to provide a more robust and diverse set of evaluations. This approach would allow for the calculation of inter-rater agreement, potentially leading to more generalizable and reliable results. It would also distribute the workload, reducing the risk of fatigue-related inconsistencies and providing a more comprehensive coverage of the various specializations within ECHR law.

8 Ethics

While our ECHR-QA benchmark is not expected to pose immediate significant risks or negative societal impacts, it is essential to consider the broader ethical implications of utilizing LLMs in legal information seeking and practice.

One primary concern is the tendency of LLMs to produce hallucinations - generating factual inaccuracies or making claims without proper support from their attributions ([Dahl et al., 2024](#)). In a legal context, where precision and accuracy are crucial, such inaccuracies could lead to serious consequences if relied upon without verification.

Additionally, LLMs have been shown to reflect and amplify biases present in their training data ([Cheong et al., 2024](#); [Gallegos et al., 2024](#); [Motoki et al., 2024](#)). This includes societal biases related to race, gender, and socioeconomic status, which may be particularly pronounced in historical legal cases ([Hennes et al., 2021](#)). The perpetuation of these biases through LLMs in legal applications could potentially lead to unfair or discriminatory outcomes.

The integration of LLMs into legal practice also raises important questions about the evolving role of legal professionals. The legal profession must maintain its integrity and high standards while incorporating these technologies. This necessitates ongoing evaluation of how LLMs can best complement legal expertise, as well as careful consideration of the ethical implications of their use in various legal contexts.

Furthermore, as LLMs become more prevalent in legal settings, there is a need for clear guidelines and ethical frameworks governing their use ([Cheong et al., 2024](#)). This includes ensuring transparency about when and how LLMs are being utilized, maintaining human oversight and accountability, and establishing protocols for verifying LLM-generated information. In conclusion, while our ECHR-QA benchmark aims to advance the field of legal NLP, it is crucial to approach the broader application of LLMs in legal domains with caution and ethical consideration. Ongoing research, open dialogue, and the development of robust ethical guidelines will be essential to harness the benefits of these technologies while mitigating potential risks and upholding the fundamental principles of justice and fairness in legal practice.

9 Future Work

This thesis opens up several avenues for future research and improvements in the field of legal question answering and the evaluation of LLMs. We can envision a wide range of potential advancements and future research directions.

In our opinion, a key area for improvement lies in developing methods for claim extraction from answers. By identifying and isolating key factors from responses, we could potentially improve the value obtained by our claim recall measure. The current sentence-based strategy leads to a claim recall of zero in most cases, while our expert found that answers often contain multiple important claims. This could also mean separating answers not just into claims, but into more granular parts, such as facts, arguments, and examples, and evaluating the performance of models on these parts.

Closely related to this is the need for metrics to assess the quality of legal argumentation in answers. Future research should focus on developing a method to evaluate the logical structure and coherence of responses, taking the specific requirements of different types of legal questions into account. This could help provide a better understanding of the effectiveness of legal QA systems.

Another remaining challenge is the development of robust methods to measure the faithfulness of citations in ECHR-QA and similar datasets. This involves not only assessing how accurately the cited passages support the claims made in the answers but also developing strategies to measure the relevance of attributions. Such an advancement could increase the reliability and trustworthiness of attributed QA systems.

Both faithfulness and correctness values could potentially be improved with novel or expanded approaches at attributed QA. Both our prompts used to generate answers (Prompts A.13, A.14) used across all strategies, do not allow additional reasoning steps (CoT), which was shown to improve performance (Wei et al., 2022). Since some of the models, especially SaulLM, had issues following our expected citation formats, enforcing some output structures¹ could be beneficial. We leave both for future work.

Further research could also investigate a wider range of retrieval strategies. We included a short experiment with BM25 (Section A.3); however, today there are more advanced embedding models, potentially leading to more effective information retrieval in legal contexts.

Interesting retrieval strategies could also involve the use of a more structured representation of ECHR judgments, such as knowledge graphs and knowledge graph embeddings (J. Liu et al., 2024; Y. Zhang et al., 2024). By representing legal concepts, cases, and their relationships in

¹For example with a library like <https://github.com/guidance-ai/guidance>

a structured format, we could potentially enhance the usefulness of the retrieved information. While ECHR-QA currently focuses only on English, future work could extend to other languages. ECHR judgments, for example, are available in both English and French. Developing multilingual datasets and models could enhance the accessibility and applicability of attributed legal QA systems, making them valuable for a broader range of legal professionals and researchers worldwide.

Although ECHR jurisprudence provides a rich area of study, it represents only a fraction of global legal contexts. Future research could use our approach to construct further attributed QA datasets covering more legal domains and jurisdictions. This expansion would allow benchmarking of legal QA systems across different legal domains and jurisdictions, providing a more comprehensive understanding of the effectiveness of generic legal QA systems and LLMs.

Future research can build upon the foundational approaches for dataset curation and evaluation proposed in ECHR-QA. We hope that our work offers insights that can be applied to legal research and education in different jurisdictions and languages, potentially leading to incremental improvements in how legal information is accessed and evaluated in the future.

A Appendix

A.1 ECHR-QA Dataset Case Study

To give the reader a better understanding of the ECHR-QA dataset, our ECHR legal expert analyzed QA pairs in the final dataset, highlighting what makes a good ECHR question-answer pair and what common issues QA pairs in the dataset face.

High-quality QA Pairs

According to our ECHR expert, a high-quality ECHR QA pair consists of a question that logically incorporates pertinent legal concepts in a manner that clearly presents a legal issue, and an answer that comprehensively addresses all elements necessary to respond to the question. The answer must include all specific elements mentioned in the question, especially those pertinent to the form of inquiry. If the legal question is “How does the law apply?” the answer should describe “How the law applies”. If the answer is about legal standards, then it should point exactly to the legal standards applicable to answer that legal question. If the question is about the conditions in which a legal notion applies, it should point to the conditions in which that legal notion applies.

QA pair 27 (see [A.1](#)) provides a good example of a golden answer. The question is about how the court protects prisoner rights and what criteria are required to determine a violation. The answer starts with the general statement and continues by describing the criteria in the Court’s case law, concluding with what needs to be considered to raise a successful complaint under Article 3. Another example of a golden QA pair is 76 (see [A.5](#)). The question is about the case of violent dispersal, where law enforcement officers use physical force. The curated answer is straightforward, starts with the general statement about human dignity, then correctly identifies the article and adds the standard supporting the application of that article to state agents (the use of force that is not strictly necessary).

Common QA Pair Issues

Our ECHR legal expert identified multiple repetitive issues during the QA pair curation as well as in the final ECHR-QA dataset.

The Model is Unable to Locate the Relevant Precedent: A common issue is the failure of the model to recognize the relevant legal issue or precedent correctly. For example,

in QA pair 43 (see [A.2](#)), the question addresses the specific issue of restrictions in prison under a particular regime in Italy. However, it also raises a broader and more fundamental question: ‘How does the Court distinguish between cases where the extension of restrictions is sufficiently justified and those where it is not, particularly in relation to an applicant’s health issues and cognitive decline?’ The answer primarily references Italian cases and fails to explain how the Court establishes the standards for restrictions comprehensively. Simply mentioning the Italian cases does not provide a full picture. To fully grasp how restrictions are applied, one would need prior knowledge of these cases and would then need to seek out additional examples to see how restrictions are applied.

Another example is QA pair 60 (see [A.3](#)), where the answer is correctly given and names the various situations that discretion in appointing a lawyer may raise, however, it fails to ground those examples in legal standards and does not link these standards to the margin of appreciation mentioned in the question. The examples that were provided by the answer raised issues under several articles of the Convention but no article or legal norm is mentioned. Unless the reader of the answer already knows the exact article and precedent the question seeks, this answer is incomplete to the reader.

The Model Fails to Identify Related Case-Law Guide Paragraphs Correctly: In many cases, the model struggles to determine how similar paragraphs should be linked, whether they are general statements or specific examples. It often fails to identify the connections between them or to understand the reasons for their differences. This issue highlights the model’s inability to ‘understand’ how a legal notion is applied differently across various situations.

For example, in QA Pair 62 of the dataset (see [A.4](#)), the wording of the question is notably imprecise. The question focuses on the regulation of audiovisual materials and the internet in prisons and in which limits can States resort to the margin of appreciation. However, the answer is in itself only partially useful, because it lacks the indication as to other factors than resources to limit that margin, even less to some examples of justification for restricting rights. In QA pair 77 (see [A.6](#)), the model-generated question concerns the circumstances under which the actions of local apparatus can be imputed to a State exercising “effective control” over an area, according to the Court’s case law. The curated answer focuses narrowly on the attribution of actions to that local apparatus similar to state agents. It is as if it assumed that one was interested only in attribution, it fails to provide the circumstances needed to justify that attribution.

The Model Fails to Address Nuances: Several issues fall under this category, some of which are language-related. The first problem concerns certain assumptions made in the answers, that may not be evident to a non ECHR expert.

For example, in QA pair 119 (see [A.7](#)), the question addresses the waiting period for accessing personal data, and the answer refers to effective remedies for accessing the data. However, the answer concludes that there is a violation of Article 8 due to an unreasonable waiting period. There is no clarification on what constitutes an unreasonable waiting period or how the remedies should be applied, which is the core of the question.

Another problem is related to the model's inability to understand the legal operational keywords, that are needed to answer a legal question. For instance, QA pair 122 (see [A.8](#)) starts by interrogating "In the context of legal recognition of gender identity, under what conditions does the Court consider it proportionate to require the conversion of a marriage into a registered partnership, and how does this relate to the State's positive obligation under Article 8 of the Convention?". Whereas QA pair 123 (see [A.9](#)) starts by interrogating "How does the Court justify the distinction between the processing of data for journalistic purposes and the dissemination of raw data in relation to Article 10 of the Convention, using the facts from the case of mass flows of personal data published in a magazine and disseminated through a text messaging service?". Both their answers started with a description of a general standard applying to the broader topic but none of them pointed to the legal techniques that are applied even less to the legal techniques through which the Court supports its conclusions. Lawyers and people wanting to present cases to the Court are looking for a specific justification or indicating that "In the case of *Hämäläinen v. Finland* [GC], 2014 (§§ 87-89), the Court considered that it had not been disproportionate to require, as a precondition to legal recognition of an acquired gender, that the applicant's marriage be converted into a registered partnership, as that was a genuine option which provided legal protection for same-sex couples that is almost identical to that of marriage does not amount to showing what lies behind this conclusion" (for QA pair 122). What is important is to emphasize the conditions that allow for such a conclusion, as this is what the question asked, but the answer does not describe how the Court develops those conditions. Similarly, the answer of QA pair 123, states that "the fact of prohibiting the mass publication of personal taxation data in a manner incompatible with Finnish and EU rules on data protection was not, as such, a sanction, despite the fact that, in practice, the limitations imposed on the quantity of the information to be published may have rendered some of the applicant companies' business activities less profitable (*ibid.*, § 197)" does not actually amount to how and why the Court has decided this way. Both examples fail to describe the required legal techniques and the legal expertise is not accounted for.

A.1.1 Example QA Pairs

Question	How does the Court's case-law protect prisoners' rights under the Convention, and what criteria does the Court use to determine if there has been a violation of those rights?
Answer	The Court has developed abundant case-law determining the nature and scope of prisoners' rights under the Convention and the duties of the domestic authorities as regards the treatment of prisoners. Prisoners in general continue to enjoy all the fundamental rights and freedoms guaranteed under the Convention save for the right to liberty, where lawfully imposed detention expressly falls within the scope of Article 5 of the Convention. For example, prisoners may not be ill-treated, subjected to inhuman or degrading punishment or conditions contrary to Article 3 of the Convention; they continue to enjoy the right to respect for family life; the right to freedom of expression; the right to practise their religion; the right of effective access to a lawyer or to a court for the purposes of Article 6; the right to respect for correspondence (<i>Hirst v. the United Kingdom</i> (no. 2) [GC], 2005, § 69). The key principle underpinning the Court's case-law related to prisoners' rights is the necessity of treatment of all persons deprived of liberty with respect for their dignity and human rights. Indeed, the very essence of the Convention system of protection of human rights is based on respect for human dignity (<i>Bouyid v. Belgium</i> [GC], 2015, §§ 89-90), which also extends to the treatment of prisoners (<i>Vinter and Others v. the United Kingdom</i> [GC], 2013, § 113). As regards the protection from violence by other prisoners, having regard to the absolute character of the protection guaranteed by Article 3 of the Convention and given its fundamental importance in the Convention system, the Court has developed a test for cases concerning a State's positive obligation under that Convention provision. In particular, it has held that to successfully argue a violation of his Article 3 right it would be sufficient for an applicant to demonstrate that the authorities had not taken all steps which could have been reasonably expected of them to prevent real and immediate risks to the applicant's physical integrity, of which the authorities had or ought to have had knowledge.

Table A.1.: QA pair 27 from ECHR-QA

Question	In the context of the special prison regime under section 41 bis, under what circumstances does the Court consider the extension of restrictions to be justified, and how does the Court differentiate between cases where the extension of restrictions is and is not sufficiently justified, with regard to an applicant's health issues and cognitive decline?
Answer	From the perspective of Article 3, the Court has held that the imposition of the 41 bis regime does not give rise of itself to an issue under Article 3, even when it has been imposed for lengthy periods of time. When assessing whether or not the extended application of certain restrictions under the section 41 bis regime attains the minimum threshold of severity required to fall within the scope of Article 3, the length of time must be examined in the light of the circumstances of each case, which entails, inter alia, ascertaining whether the renewal or extension of the impugned restrictions was justified or not (<i>Provenzano v. Italy</i> , 2018, § 147, with further references). Thus, for instance, in <i>Enea v. Italy</i> [GC], 2009, §§ 60-67, the Court considered that the restrictions imposed as a result of the special prison regime were necessary to prevent the applicant, who posed a danger to society, from maintaining contacts with the criminal organisation to which he belonged. Thus, irrespective of the health issues from which the applicant suffered, the Court found no violation of Article 3 of the Convention. By contrast, in <i>Provenzano v. Italy</i> , 2018, §§ 149-158, the Court considered that the extension of the application of the 41 bis regime in respect of the applicant had not been sufficiently justified, particularly having regard to his critical cognitive decline.

Table A.2.: QA pair 43 from ECHR-QA

Question	How does the margin of appreciation apply in relation to the State's discretion to appoint a lawyer for an applicant under a legal-aid scheme and ensure their professional fulfillment of duties, as required by the applicant's right to individual application?
Answer	Similarly, in some instances, the exercise of an applicant's right to individual application in a concrete and effective manner may imply the necessity for the respondent State to undertake positive measures, such as appointing a lawyer to the applicant under a legal-aid scheme and ensuring that the thus appointed lawyer fulfils the duties in a professional manner and to the best of his/her ability (<i>Feilazoo v. Malta</i> , 2021, §§ 125-132). Prompt access to a lawyer constitutes an important counterweight to the vulnerability of persons in custody, provides a fundamental safeguard against coercion and ill-treatment, and contributes to the prevention of miscarriages of justice and the fulfilment of the aims of a fair trial under Article 6 of the Convention (<i>Salduz v. Turkey</i> [GC], 2008, §§ 53-54; <i>Ibrahim and Others v. the United Kingdom</i> [GC], 2016, § 255).

Table A.3.: QA pair 60 from ECHR-QA

A. Appendix

Question	Based on the given facts, and considering the criteria set forth by the Court, has the Contracting State's margin of appreciation been respected in regulating the access to audiovisual training materials, computers, and the Internet in prisons?
Answer	On the merits of the case, the Court found that the manner of regulating access to audiovisual training materials, computers and the Internet fell within the Contracting State's margin of appreciation. The prisons in the present case had the means to provide their inmates with the possibility afforded by the law. Moreover, no concrete justification for the lack of resources of the prisons in question had been put forward in the domestic proceedings or before the Court nor did the domestic authorities conduct an appropriate assessment of all the interests at stake or provide any justification for restricting the applicants' access to the use of the electronic material (<i>Ibid.</i> , §§ 60-72).

Table A.4.: QA pair 62 from ECHR-QA

Question	In the case of a violent dispersal of an assembly, where law-enforcement officers used physical force against individuals even when it was not strictly necessary based on their conduct, which Article of the Convention is applicable?
Answer	The Court stressed that any interference with human dignity strikes at the very essence of the Convention. Therefore, any conduct by law-enforcement officers vis-à-vis an individual which diminishes human dignity constitutes a violation of Article 3. That applies in particular to their use of physical force against an individual where it is not made strictly necessary by his or her conduct, whatever the impact on the person in question (<i>Bouyid v. Belgium</i> [GC], 2015, § 101). The Court has applied these principles also in cases concerning violent dispersal of assemblies (<i>Navalnyy and Gunko v. Russia</i> , 2020, §§ 41 and 48; <i>Zakharov and Varzhabetyan v. Russia</i> , 2020, §§ 62 and 71).

Table A.5.: QA pair 76 from ECHR-QA

Question	Under what circumstances can the actions of local apparatus be imputed to a State exercising "effective control" of an area, according to the Court's case-law?
Answer	Lastly, it should be noted that according to the Court's case-law, not only actions of State agents could be imputed to the Contracting State, but also the actions of local apparatus can be imputable to the State exercising "effective control" of the area (<i>Djavit An v. Turkey</i> , 2003, § 22).

Table A.6.: QA pair 77 from ECHR-QA

A. Appendix

Question	How does the Court determine whether the waiting period for accessing personal data constitutes a violation of Article 8?
Answer	The effectiveness of remedies available at the domestic level for persons wishing to have access to their personal data requires applications submitted by the data subjects to be processed within a reasonable time. In the case of <i>Roche v. the United Kingdom</i> [GC], 2005 (§§ 166-167, 169), the Court found a violation of Article 8 on account of an unreasonable waiting period for the applicant in accessing documents comprising personal data which would have enabled him to assess the potential risks to his health caused by his participation in military testing on gases.

Table A.7.: QA pair 119 from ECHR-QA

Question	In the context of legal recognition of gender identity, under what conditions does the Court consider it proportionate to require the conversion of a marriage into a registered partnership, and how does this relate to the State's positive obligation under Article 8 of the Convention?
Answer	While the boundaries between the State's positive and negative obligations under the Convention do not lend themselves to precise definition, the applicable principles are nonetheless similar. In both contexts regard must be had in particular to the fair balance that has to be struck between the competing interests of the individual and of the community as a whole, subject in any event to the margin of appreciation enjoyed by the State (<i>Bărbulescu v. Romania</i> [GC], 2017, § 112). In the case of <i>Hämäläinen v. Finland</i> [GC], 2014 (§§ 87-89), the Court considered that it had not been disproportionate to require, as a precondition to legal recognition of an acquired gender, that the applicant's marriage be converted into a registered partnership, as that was a genuine option which provided legal protection for same-sex couples that is almost identical to that of marriage (homosexual marriage being illegal in Finland). Consequently, the minor differences between these two legal concepts were not such as to render the current Finnish system deficient from the point of view of the State's positive obligation.

Table A.8.: QA pair 122 from ECHR-QA

Question	How does the Court justify the distinction between the processing of data for journalistic purposes and the dissemination of raw data in relation to Article 10 of the Convention, using the facts from the case of mass flows of personal data published in a magazine and disseminated through a text messaging service?
Answer	In the Court's view, the existence of a public interest in providing access to, and allowing the collection of, large amounts of taxation data for journalistic purposes did not necessarily or automatically mean that there was also a public interest in disseminating en masse such raw data in unaltered form without any analytical input. A distinction had to be made between the processing of data for journalistic purposes and the dissemination of the raw data to which the journalists were given privileged access (ibid., § 175). In that context, the fact of prohibiting the mass publication of personal taxation data in a manner incompatible with Finnish and EU rules on data protection was not, as such, a sanction, despite the fact that, in practice, the limitations imposed on the quantity of the information to be published may have rendered some of the applicant companies' business activities less profitable (ibid., § 197).

Table A.9.: QA pair 123 from ECHR-QA

Question	How does the Court assess the imputability of impugned acts to the State in cases where the behavior of a State agent is unlawful?				
Answer	The Court has held that where the behavior of a State agent is unlawful, the question of whether the impugned acts can be imputed to the State requires an assessment of the totality of the circumstances and consideration of the nature and circumstances of the conduct in question. Moreover, whether a person is an agent of the State for the purposes of the Convention is defined on the basis of a multitude of factors, none of which is determinative on its own. The key criteria used to determine whether the State is responsible for the acts of a person, whether formally a public official or not, are as follows: manner of appointment, supervision and accountability, objectives, powers and functions of the person in question (V.K. v. Russia, 2017, § 174).				
Citations	V.K. v. Russia, 2017, § 174: A Contracting State will be responsible under the Convention for violations of human rights caused by acts of its agents carried out in the performance of their duties. The Court has held that where the behaviour of a State agent is unlawful, the question of whether the impugned acts can be imputed to the State requires an assessment of the totality of the circumstances and consideration of the nature and circumstances of the conduct in question (see Reilly v. Ireland (dec.), no. 51083/09, § 53, 23 September 2014, with further references).				
Evaluation	Q. Relevance	Q. Fluency	A. Fluency	A. Compr.	A. Concise.
	3	5	5	4	5

Table A.10.: Example expert annotated QA pair generated with Legal Sentence-Level-CoT with Search v1 for Guide Art 3

A.2 Attributed QA Case Study

We present several examples to examine the characteristics of the generated responses. The examples were selected and analyzed in collaboration with our legal expert.

Example 1: Comparison of Strategies

A.11 provides an ECHR-QA question related to the open-ended topic of data protection, the target answer and responses from the Mistral-7B model including answers from the base strategy (closed book), as well as RAG and LLaTrieval.

The base answer delineates general principles relevant to the disclosure of electronic information, elaborating on a range of scenarios governed by these principles. However, it fails to provide a precise answer to the core question regarding the defense's involvement in setting criteria for judicial proceedings. The RAG approach supplements the base answer by detailing specific circumstances related to the defense's role in defining disclosure criteria, such as ensuring that the individuals' ability to defend themselves is not impeded by disclosure. Nevertheless, RAG lacks precise information about safeguards against arbitrariness in cases requiring a balance of interests.

LLaTrieval provides concise text but does not offer new insights into the defense's role, which is central to the question. It also overlooks specifying problematic data aspects for the defense and the disclosure of the issue may raise, unlike RAG. Both models rightly emphasize the defense's need to challenge, impacting procedure quality. While LLaTrieval tends towards more thorough arguments, vanilla RAG leans towards general principles.

In terms of language use, the base answer employs less legal jargon. For example, it uses non-legal language such as "help streamline the disclosure" which would translate to "provides additional guarantees" in legal terms. In contrast, the retrieval approach incorporates the legal language from the presented case-law documents. Retrieval also aids logical steps, reflected in the overall reasoning of the retrieval-based approaches.

Example 2: Comparison of RAG across Models

A.12 provides an ECHR-QA question addressing the legal issue of determining the applicable Article to a situation where a foreign prisoner is denied contact with their family, along with the target answer and vanilla RAG responses from our four different LLMs.

An ideal response would specify that Article 8 is applicable because maintaining contact with family while in prison has been ruled as part of a prisoner's private life, guaranteed under that Article. The Court analyzes each limitation on access to the family as an interference, and this interference is then assessed in light of various considerations that the Court deems relevant to the specific situation, forming the core of the Court's reasoning. There are common considerations that apply to all Articles of the Convention, as well as specific considerations unique to each Article. Specific considerations for Article 8 in dealing with the right of prisoners to have contact with their family and access to the outside world are legislation, practical conditions of transportation, geographic location, and possibilities for the prisoner's family to

visit. In addition, when analyzing restrictions on prisoners, the Court has established that States have an obligation to assist the detainee in keeping contact with his family.

Generally, for this type of question, the greater the number of considerations as to why Article 8 is a candidate, the better the support to justify the application of that Article. Whenever lawyers and judges look for reasons justifying the application of an Article to a new situation they validate already existing considerations applicable to a situation in order to check whether these considerations may or may not apply to their situation.

All models failed to provide comprehensive details of all involved legal aspects. Aspects the Court would consider, such as conjugal visits and access to social media, were not mentioned. Mistral provided the best answer, offering a brief overview of why Article 8 applies to the given situation by systematically going through all the steps of the Court's core reasoning structure (essential aspect hardship, interests, normal hardships, and restrictions) and identifying the most relevant circumstances influencing that reasoning (geographical location, interests of both the prisoner and his family, distance, and financial hardship).

Llama-3-8b includes general statements of why Article 8 applies, but only briefly mentions a few special circumstances at the end (like a prisoner's young age and financial hardship) before concluding abruptly on the possible application. Llama-3-70B's response was overall better than Llama-3-8B's even though Llama-3-8B had additional details such as whether the prisoner has young children and financial hardships. This is primarily because it concludes that the Court will likely analyze whether the effects on private life go beyond the normal hardships and restrictions inherent in the very concept of imprisonment. Llama-3-8B, on the other hand, concludes that the Court would likely find the denial of any contact with the family to be a disproportionate interference, thereby providing a more straightforward answer. It gives the impression that this is what the Court would conclude, but in determining whether the interference is disproportionate, the Court first analyzes circumstances that go beyond or fall below the normal hardships, involving a series of sequential reasoning steps.

The legally pre-trained SaulLM uses specific legal terminology but presents the conclusion as a given. It offers general statements about what constitutes a hardship but fails to explain why this would always amount to a violation in the case of denial of access. Legal professionals typically anticipate pros and cons for hypotheses rather than straightforward answers, especially for a question like this. SaulLM skipped the intermediary steps of the decision-making process. One has to strike a balance between general statements and principles and the specific circumstances of the question to make it legally informative and to be perceived as useful by a legal professional. While SaulLM employs legalese, its rhetoric is more imposing than argumentative. Understanding the nuances of the decision process is crucial.

Example 3: Edits by RARR

To highlight the potential utility of the editing phenomenon in RARR, we showcase examples where the model performed edits. We present the question with the response from the closed-book approach including edits from RARR highlighting the sentences that underwent an edit.

In the context of the question provided in [A.13](#), Article 12 applies to transgender individuals

wishing to marry a person of the opposite sex (i.e., opposite to their newly assigned sex), as well as to same-sex couples wishing to marry or who are already married. However, only a total ban on the former constitutes a violation of Article 12, and a total ban on the latter is to date Convention compliant. We observe that RARR using SaulLM correctly identifies the wrongly presented information and successfully edits it to make it faithful and accurate.

The second example (A.14) showcases a response from Llama-3-8B, where the model makes two edits. The first edit appears faithful and accurate, as the base model states that revocation is proportionate to the legitimate aim of maintaining trust and safety. However, restrictions affecting a professional license are typically considered in light of the applicant's rights and reputation, rather than solely in relation to that aim. The second edit is a stylistic change. Not finding supporting evidence and finding only citations labeled as disagreements, the model is forced to edit but correctly preserves the original content.

A.15 presents another response from Llama-3-8B. The first two edits do not introduce new information but rephrase the base text, similar to the second edit in the previous example. The last edit states the contrary to the base, but also contrary to the decision of the Court and the retrieved text. This highlights a vulnerability of the model and strategy in which unfaithful or wrong content is introduced. The Court concluded that the restrictions were not proportionate to the aim while the edit claims the contrary. Moreover, the base correctly says that measures lacked clear criteria for application and were arbitrary. However, the edit states that measures follow the law and have no impact on the right to private life, which stands completely inaccurate.

A.2.1 Example Generations

Question: In the context of disclosure of electronic data, what criteria should be considered to determine what might be relevant for disclosure, and why is the defense's involvement in laying down these criteria important?
Target Answer: In the context of disclosure of evidence, complex issues may arise concerning the disclosure of electronic data, which may constitute a certain mass of information in hands of the prosecution. In such a case, an important safeguard in the sifting process is to ensure that the defence is provided with an opportunity to be involved in the laying-down of the criteria for determining what might be relevant for disclosure (Sigurður Einarsson and Others v. Iceland, 2019, § 90; see also Rook v. Germany, §§ 67 and 72).
Base Answer: In the context of disclosure of electronic data before the European Court of Human Rights (ECtHR) or any other court, the relevance of the data is a key consideration. Relevance refers to the data's ability to make a fact more or less probable as to an allegation or issue in the case. When it comes to electronic data, several criteria should be considered to determine its relevance for disclosure: 1. Proximity: Is there a logical connection between the data and the facts or issues at hand? For example, emails or messages exchanged between parties may be highly relevant if they discuss the alleged misconduct or contain important information. 2. Authenticity: Does the data origin from a trusted or reliable source? Has its integrity been maintained during collection and storage? The authenticity of electronic data is crucial for it to be considered admissible and relevant. 3. Admissibility: Does the data possess probative value, or can it be used to prove or disprove a particular fact? For example, text messages, emails, or social media postings can be admissible as evidence provided they meet the authenticity and relevance requirements. 4. Quantity: Is the volume of data manageable and worth the resources required for its processing? The sheer amount of digital data can sometimes be overwhelming, and it should not be disproportionate to the issue at hand or result in an undue burden on the parties involved. 5. Format: Is the format of the data easily accessible and understandable to the court, as well as the parties involved? This criterion is crucial since data can be available in various formats, making it imperative to understand the capabilities and limitations of each format and the means required to access and interpret it. The defense's involvement in laying down these criteria is essential for several reasons: 1. Protection of their rights: The defense has a fundamental right to a fair trial and must be provided with evidence that could reasonably be expected to aid their defense. Therefore, they have a vested interest in ensuring that the appropriate criteria for determining relevance and admissibility of electronic data are met. 2. Efficiency: The defense's input can help streamline the disclosure process by focusing on data that is truly relevant and admissible. This ensures that the trial does not become unnecessarily prolonged, saving resources and time for all parties. 3. Transparency: The defense's input in the disclosure process can promote transparency and ensure that all evidence considered in the case is accessible and understandable to all parties. This is crucial to maintaining the integrity and impartiality of the judicial process.
RAG: In the context of disclosure of electronic data, the criteria to determine what might be relevant for disclosure include evidence directly relevant to the facts of the case and other evidence that might relate to the admissibility, reliability, and completeness of the former (CASE OF YAKUBA v. UKRAINE § 43, CASE OF MATANOVIĆ v. CROATIA § 161). The material the disclosure of which the applicant sought may fall into the latter category (CASE OF YEVDOKIMOV v. UKRAINE § 46). The defense's involvement in laying down these criteria is important as they can ensure that the impact of the relevant material on the safety of the conviction is considered in the light of detailed and informed arguments (CASE OF MATANOVIĆ v. CROATIA § 155). The Court must satisfy itself that the domestic procedure allows for the consideration of the relevance and impact of the undisclosed material in an effective manner. Furthermore, in the context of communications data, it should be noted that these data are often subjected to automated processing and queries, and applying the same safeguards as for content is not practicable due to the large volume and frequent temporal nature of the data (CASE OF BIG BROTHER WATCH AND OTHERS v. THE UNITED KINGDOM § 296). However, the defense's role in laying down these criteria is crucial in ensuring that any limitations on the disclosure of communications data do not compromise the requirements of Article 6. Overall, the defense's involvement in determining what is relevant for disclosure is essential in ensuring the fairness and transparency of criminal proceedings, as they can help ensure that all material evidence is disclosed to the defense, enabling them to effectively challenge the prosecution's case and protect their client's rights (CASE OF YAKUBA v. UKRAINE § 43).
LLatrieval: In the context of disclosure of electronic data, the criteria that should be considered to determine what might be relevant for disclosure include the opportunity for the defense to challenge the authenticity of the evidence and to oppose its use (CASE OF PAVLENKO v. RUSSIA § 116, CASE OF ILGAR MAMMADOV v. AZERBAIJAN (No. 2) § 209, CASE OF GÄFGEN v. GERMANY § 95, CASE OF VUKOTA-BOJIC v. SWITZERLAND § 95, CASE OF SVETINA v. SLOVENIA § 44). The quality of the evidence obtained, including the circumstances in which it was obtained, must also be taken into consideration, as these factors may cast doubts on its reliability or accuracy (CASE OF PAVLENKO v. RUSSIA § 116, CASE OF ILGAR MAMMADOV v. AZERBAIJAN (No. 2) § 209, CASE OF GÄFGEN v. GERMANY § 95, CASE OF VUKOTA-BOJIC v. SWITZERLAND § 95, CASE OF SVETINA v. SLOVENIA § 44). The Court has previously held that where the reliability of evidence is in dispute, the existence of fair procedures to examine its admissibility takes on an even greater importance (CASE OF PAVLENKO v. RUSSIA § 116). The defense's involvement in laying down these criteria is important because it ensures that the rights of the defense are respected throughout the proceedings (CASE OF PAVLENKO v. RUSSIA § 116, CASE OF ILGAR MAMMADOV v. AZERBAIJAN (No. 2) § 209, CASE OF GÄFGEN v. GERMANY § 95, CASE OF VUKOTA-BOJIC v. SWITZERLAND § 95, CASE OF SVETINA v. SLOVENIA § 44). The quality of the electronic data and the circumstances in which it was obtained are also crucial factors that the defense should be allowed to examine and challenge, as they may impact the reliability and accuracy of the evidence (CASE OF PAVLENKO v. RUSSIA § 116, CASE OF ILGAR MAMMADOV v. AZERBAIJAN (No. 2) § 209, CASE OF GÄFGEN v. GERMANY § 95, CASE OF VUKOTA-BOJIC v. SWITZERLAND § 95, CASE OF SVETINA v. SLOVENIA § 44). By involving the defense in the determination of what is relevant for disclosure, the fairness of the proceedings as a whole is upheld.

Table A.11.: Example Mistral Answers

A. Appendix

Question: Which article of the Convention would be applicable to a situation where a foreign prisoner is denied any contact with their family, including through telephone conversations or occasional visits, and what reasons would the Court provide to justify the application of that article to these facts?
Target Answer: As the Convention does not guarantee as such the right to an inter-state transfer or the right of a detainee to be allocated to a particular prison, it is important to ensure that foreign prisoners maintain some contact with their families, at least through telephone conversations or occasional visits (Labaca Larrea and Others v. as. France (dec.), 2017, § 44). Moreover, the authorities may be required under Article 8 to make concessions for allowing a prisoner to contact and speak to his or her family members in their own language (Nusret Kaya and Others v. Turkey, 2014, §§ 60-61).
Llama 8b: Based on the facts provided in the paragraphs, the applicable article of the Convention would be Article 8, which guarantees the right to respect for private and family life. This is because the prisoner's denial of any contact with their family, including through telephone conversations or occasional visits, would likely have a significant impact on their private and family life. The Court would likely justify the application of Article 8 to these facts by highlighting the importance of maintaining contact with one's family, particularly in the context of imprisonment. The Court has already established that it is an essential part of a prisoner's right to respect for family life that the authorities enable him or her, or if need be, assist him or her, to maintain contact with his or her close family (CASE OF YUDIN AND OTHERS v. RUSSIA § 3, CASE OF RAZVOZZHAYEV v. RUSSIA AND UKRAINE AND UDALTSOV v. RUSSIA § 270, CASE OF VOYNOV v. RUSSIA § 49, CASE OF POLYAKOVA AND OTHERS v. RUSSIA § 81, CASE OF SAIDOV AND OTHERS v. RUSSIA § 7, CASE OF KHAMASTKHANOVA AND OTHERS v. RUSSIA § 10, CASE OF YEVSYUKOV AND OTHERS v. RUSSIA § 7). The Court would likely emphasize that the denial of any contact with the family would go beyond the "normal" hardships and restrictions inherent in the very concept of imprisonment, particularly in view of the geographical situation of remote penal facilities and the realities of the transport system (CASE OF RAZVOZZHAYEV v. RUSSIA AND UKRAINE AND UDALTSOV v. RUSSIA § 270, CASE OF POLYAKOVA AND OTHERS v. RUSSIA § 81, CASE OF SAIDOV AND OTHERS v. RUSSIA § 7, CASE OF KHAMASTKHANOVA AND OTHERS v. RUSSIA § 10, CASE OF YEVSYUKOV AND OTHERS v. RUSSIA § 7). In addition, the Court would likely consider the specific circumstances of the case, such as the prisoner's young child and the financial hardship of the relatives, to justify the application of Article 8 (CASE OF DANILEVICH v. RUSSIA § 9). The Court would likely conclude that the denial of any contact with the family would be a disproportionate interference with the prisoner's right to respect for private and family life, and would therefore be in violation of Article 8 of the Convention.
Llama 70b: The applicable article of the Convention would be Article 8, which guarantees the right to respect for private and family life (CASE OF YUDIN AND OTHERS v. RUSSIA § 3). The Court would justify the application of Article 8 to these facts by stating that it is an essential part of a prisoner's right to respect for family life that the authorities enable him or her, or if need be assist him or her, to maintain contact with his or her close family (CASE OF RAZVOZZHAYEV v. RUSSIA AND UKRAINE AND UDALTSOV v. RUSSIA § 270, CASE OF VOYNOV v. RUSSIA § 49, CASE OF POLYAKOVA AND OTHERS v. RUSSIA § 81, CASE OF SAIDOV AND OTHERS v. RUSSIA § 7, CASE OF KHAMASTKHANOVA AND OTHERS v. RUSSIA § 10, CASE OF YEVSYUKOV AND OTHERS v. RUSSIA § 7). The Court would also emphasize that, on the issue of family visits, Article 8 of the Convention requires States to take into account the interests of the prisoner and his or her relatives and family members (CASE OF RAZVOZZHAYEV v. RUSSIA AND UKRAINE AND UDALTSOV v. RUSSIA § 270, CASE OF VOYNOV v. RUSSIA § 49, CASE OF POLYAKOVA AND OTHERS v. RUSSIA § 81, CASE OF SAIDOV AND OTHERS v. RUSSIA § 7, CASE OF KHAMASTKHANOVA AND OTHERS v. RUSSIA § 10, CASE OF YEVSYUKOV AND OTHERS v. RUSSIA § 7). Furthermore, the Court would consider that placing a prisoner in a particular penal facility may raise an issue under Article 8 of the Convention if its effects on his or her private and family life go beyond the "normal" hardships and restrictions inherent in the very concept of imprisonment (CASE OF RAZVOZZHAYEV v. RUSSIA AND UKRAINE AND UDALTSOV v. RUSSIA § 270, CASE OF VOYNOV v. RUSSIA § 49, CASE OF POLYAKOVA AND OTHERS v. RUSSIA § 81, CASE OF SAIDOV AND OTHERS v. RUSSIA § 7, CASE OF KHAMASTKHANOVA AND OTHERS v. RUSSIA § 10, CASE OF YEVSYUKOV AND OTHERS v. RUSSIA § 7).
Mistral: The applicable article of the Convention would be Article 8, which guarantees the right to respect for private and family life (CASE OF YUDIN AND OTHERS v. RUSSIA § 3). The Court would justify the application of Article 8 to these facts by stating that it is an essential part of a prisoner's right to respect for family life that the authorities enable him or her, or if need be assist him or her, to maintain contact with his or her close family (CASE OF RAZVOZZHAYEV v. RUSSIA AND UKRAINE AND UDALTSOV v. RUSSIA § 270, CASE OF VOYNOV v. RUSSIA § 49, CASE OF POLYAKOVA AND OTHERS v. RUSSIA § 81, CASE OF SAIDOV AND OTHERS v. RUSSIA § 7, CASE OF KHAMASTKHANOVA AND OTHERS v. RUSSIA § 10, CASE OF YEVSYUKOV AND OTHERS v. RUSSIA § 7). The Court would also emphasize that, on the issue of family visits, Article 8 of the Convention requires States to take into account the interests of the prisoner and his or her relatives and family members (CASE OF RAZVOZZHAYEV v. RUSSIA AND UKRAINE AND UDALTSOV v. RUSSIA § 270, CASE OF VOYNOV v. RUSSIA § 49, CASE OF POLYAKOVA AND OTHERS v. RUSSIA § 81, CASE OF SAIDOV AND OTHERS v. RUSSIA § 7, CASE OF KHAMASTKHANOVA AND OTHERS v. RUSSIA § 10, CASE OF YEVSYUKOV AND OTHERS v. RUSSIA § 7). Furthermore, the Court would consider that placing a prisoner in a particular penal facility may raise an issue under Article 8 of the Convention if its effects on his or her private and family life go beyond the "normal" hardships and restrictions inherent in the very concept of imprisonment (CASE OF RAZVOZZHAYEV v. RUSSIA AND UKRAINE AND UDALTSOV v. RUSSIA § 270, CASE OF VOYNOV v. RUSSIA § 49, CASE OF POLYAKOVA AND OTHERS v. RUSSIA § 81, CASE OF SAIDOV AND OTHERS v. RUSSIA § 7, CASE OF KHAMASTKHANOVA AND OTHERS v. RUSSIA § 10, CASE OF YEVSYUKOV AND OTHERS v. RUSSIA § 7).
Saul: The Convention article that would be applicable to a situation where a foreign prisoner is denied any contact with their family, including through telephone conversations or occasional visits, would be Article 8 (CASE OF POLYAKOVA AND OTHERS v. RUSSIA § 81, CASE OF SAIDOV AND OTHERS v. RUSSIA § 7). The Court justifies the application of Article 8 to these facts because it establishes that it is an essential part of a prisoner's right to respect for family life that the authorities enable them, or if need be, assist them, to maintain contact with their close family.

Table A.12.: Example RAG Answers

A. Appendix

Question: How does the Court differentiate between the rights of transgender individuals and same-sex couples in relation to the right to marry under Article 12, and what justifies the distinction between a total ban on marriage for transgender individuals and a total ban on same-sex marriage?

Answer: The European Court of Human Rights (ECtHR) acknowledges that the rights of transgender individuals and same-sex couples are distinct under the ambit of Article 12, which provides the right to marry. In evaluating the rights of transgender individuals, the ECtHR understands that their legal status is that of a person of the sex assigned at birth and their requests to get married relate to their acquired gender. **<base>**In contrast, same-sex couples' requests to marry involve the right to form a different-sex marriage, which is not their legal status.**</base>** **<edit>**Same-sex couples' requests to marry involve the right to form a same-sex marriage, which is not their legal status.**</edit>** The Court also differentiates between the circumstances surrounding the two groups' right to marry. Transgender individuals may suffer from legal impediments such as statutory bans on marriages between persons of the same sex, making it challenging to marry after undergoing gender reassignment surgery. In contrast, same-sex couples do not face legal obstacles in most European States when they try to get married, and their request is to change the legal description of marriage rather than facilitate marriage for the same-sex couple. Furthermore, the ECtHR recognizes the difference in the justifications behind the ban on marriage for transgender individuals and a ban on same-sex marriage. A ban on marriage for transgender individuals, specifically when a transgender person has undergone gender reassignment surgery, infringes on the right to marry under Article 12, as it hinders their ability to marry. In this context, a total ban on marriage of transgender individuals is discriminatory and unlawful. On the other hand, a total ban on same-sex marriage stems from the aim of protecting the traditional institution of marriage being established between a man and a woman (e.g. Obergefell, cited above). The ECtHR acknowledges that this difference in justifications means that while it might be acceptable to impose a total ban on same-sex marriage for the protection of the traditional form of marriage, it is unlawful to impose a total ban on transgender individuals' right to marry due to the disparate nature of the justifications behind the restrictions.

Table A.13.: Example Edit by Saul using Post-hoc Retrieval - RARR

Question: In a case involving the revocation of a professional license due to pending criminal proceedings, under what conditions would the Court find that a State has failed to fulfill its positive obligations under Article 8 of the Convention?

Answer: A great question! To determine whether a State has failed to fulfill its positive obligations under Article 8 of the European Convention on Human Rights, we need to carefully examine the circumstances of the case. Article 8 of the Convention, which protects the right to respect for private and family life, has an "interface" with the concept of a professional license, which is essential for a person's livelihood and career. The Court has developed a nuanced approach to assess whether a State has breached its positive obligations under Article 8 when revoking a professional license due to pending criminal proceedings. Here are the key conditions:

1. ****Interference with the right to respect for private and family life****: The Court will first assess whether the revocation of the professional license interferes with the individual's right to respect for private and family life, including their professional life. This may involve considering the gravity of the pending criminal charges and the impact of the revocation on the individual's ability to maintain their livelihood and contribute to their family's well-being.
2. ****Proportionality****: **<base>**The Court will then consider whether the revocation of the professional license is proportionate to the legitimate aim sought to be achieved, such as maintaining public trust and safety.**</base>** **<edit>**The Court will then consider whether the revocation of the professional license is proportionate to the legitimate aim sought to be achieved, such as protecting the applicant's private life and reputation.**</edit>**
3. ****Alternative measures****: The Court will evaluate the relationship between the measure and the aim, considering factors like the seriousness of the criminal charges, the individual's professional track record, and the potential risks to the public.**</base>** **<edit>**The Court will evaluate the proportionality of the measure, considering factors like the seriousness of the criminal charges, the individual's professional track record, and the potential risks to the public.**</edit>**
4. ****Reasonable relationship****: The Court will assess whether there is a reasonable relationship between the measures taken and the legitimate aim sought to be achieved. The measures should be suitable, necessary, and proportionate to the aim.

If the State fails to demonstrate that its measures are reasonable and proportionate, and that alternative measures were not available or considered, the Court may find that the State has failed to fulfill its positive obligations under Article 8. In conclusion, for the Court to find that a State has failed to fulfill its positive obligations under Article 8, the State's measures to revoke a professional license due to pending criminal proceedings must be disproportionate, unreasonable, and lacking in proportionality. The Court will scrutinize the measures taken, considering the individual's right to respect for private and family life, the proportionality of the measures, and the availability of alternative measures.

Table A.14.: Example Edit by Llama 8b using Post-hoc Retrieval - RARR

A. Appendix

Question: In the context of employment restrictions, why did the Court find that the imposition of such restrictions on former KGB employees in the private sector cannot be justified under Article 14 of the Convention taken in conjunction with Article 8?
Answer: A great question! In the landmark case of E.B. v. France, the European Court of Human Rights (ECtHR) ruled that the imposition of employment restrictions on former KGB employees in the private sector cannot be justified under Article 14 of the Convention taken in conjunction with Article 8. In essence, the Court held that the restrictions violated the applicants' right to respect for their private and family life (Article 8) and their rights under the non-discrimination principle (Article 14). Here's why: 1. Lack of proportionality: The restrictions were deemed to be disproportionate to the legitimate aim of preventing the dissemination of classified information. The Court found that the sanctions were too severe and did not take into account the applicants' individual circumstances or the nature of their work. 2. Arbitrary and unpredictable application: <base>The Court criticized the lack of clear and stable criteria for applying the employment restrictions.</base> <edit>The Court criticized the lack of a clear legal basis for the domestic authorities' decisions in applying the employment restrictions.</edit> <base>This led to arbitrary and unpredictable decision-making, which undermined the applicants' trust in the system.</base> <edit>The Court criticized the lack of clear and transparent decision-making, which undermined the applicants' trust in the system.</edit> 3. Disproportionate impact on certain groups: The restrictions disproportionately affected certain groups, such as former KGB employees in the private sector, who had already undergone strict screening and had been vetted by their employers. The Court found that these individuals were being punished twice for the same "offense," first during their time as government employees and again in their private sector careers. 4. No effective domestic remedy: The applicants had no effective domestic remedy to challenge the employment restrictions. The Court held that the applicants' right to an effective remedy (Article 13) was also violated. In conclusion, the ECtHR ruled that the employment restrictions on former KGB employees in the private sector were not justified under Article 14 taken in conjunction with Article 8 because they were arbitrary, disproportionate, and lacked clear criteria for application. <base>The restrictions were deemed to be a disproportionate interference with the applicants' rights, including their right to respect for their private and family life.</base> <edit>The restrictions were deemed to be in accordance with the law and necessary in a democratic society, including their impact on the applicants' right to respect for their private and family life.</edit>

Table A.15.: Example Edit by Llama 8b using Post-hoc Retrieval - RARR

A.3 BM25

To see if values would greatly vary when using BM25 instead of GTR we conducted a small experiment with RAG and Llama-3-70B for the first 100 QA pairs of ECHR-QA. This small experiment suggests that BM25 might be slightly better than GTR for the ECHR-QA dataset. Especially the evidence EM is higher for BM25. However, we do not expect that the (additional) use of BM25 would lead to significantly different results. We leave the use of BM25 and other retrieval strategies for future work.

	Rouge-L	BERTScore	Claim Rec.	Citation Rec.	Evidence EM	Evidence Rec.
GTR	25.6	60.4	8.3	71.1	8.4	49.3
BM25	27.9	61.9	8.7	64.7	12.4	46.7

Table A.16.: GTR vs. BM25 for RAG Llama-3-70B

A.4 Figures

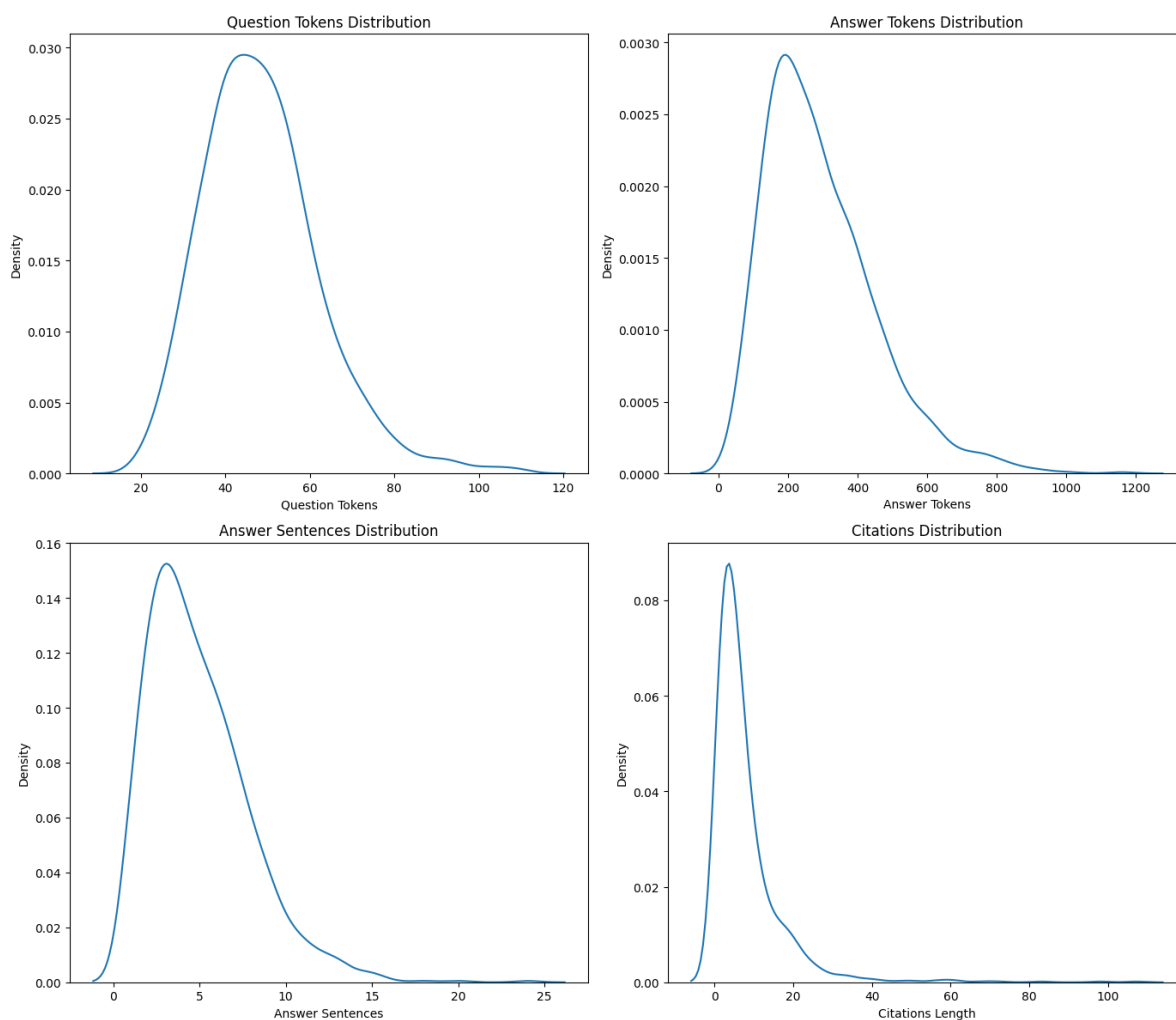


Figure A.1.: ECHR-QA Data Analysis

A. Appendix

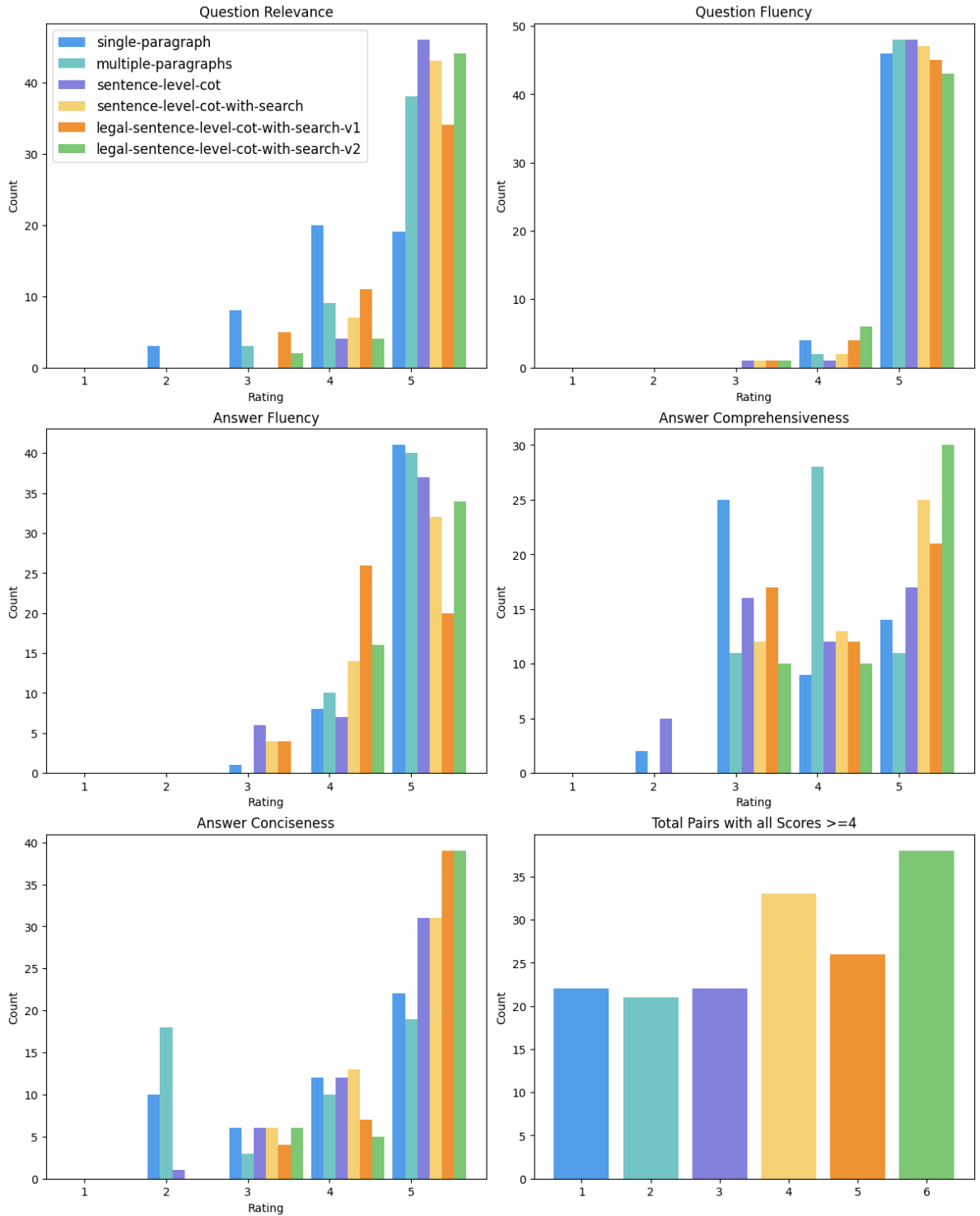


Figure A.2.: Expert QA Pair Evaluation Distributions

A.5 ECHR-QA QA Pair Curation Prompts

This section presents the prompts used during our QA pair curation strategies presented in Chapter 3 [ECHR-QA Dataset Generation](#).

The *Single-Paragraph* prompt [A.3](#) generates a question based on a single {PARAGRAPH} from the case law guides. The provided paragraph serves as the answer to the generated question. The *Multiple-Paragraphs* prompt [A.4](#) generates a question based on 5 sequentially selected paragraphs ({PARAGRAPHS[5]}) pretended with enumerations [i]: used as separators and as identifiers for the selected paragraphs in the response which we extract via the answer template.

The *Sentence-Level-CoT* prompt [A.5](#) again generates a question based on 5 sequentially selected paragraphs. The placeholder {PARAGRAPH_SENTENCES[5]} however includes k sentence level enumerations [i]: such that the model can curate the answer on a sentence level instead of at the paragraph level.

We have two *Sentence-Level-CoT with Search* prompts. Question generation prompt [A.6](#) includes 3 paragraphs and generates a question based on them. After we retrieve the top 5 paragraphs using embedding search with the generated question across all the paragraphs in the respective case-law guide, we task the LLM to generate an answer with sentence-level citations with prompt [A.7](#). Prompt [A.7](#) includes the union of the 3 original paragraphs and the 5 retrieved paragraphs with sentence-level enumerations and the {QUESTION} to generate an answer with sentence level citations. We then use these citations to curate the final answer. Our *Legal Sentence-Level-CoT with Search v1* follows the same strategy only with redefined prompts [A.8](#) and [A.10](#) for question and answer generation. *Legal Sentence-Level-CoT with Search v2* only updates the question-generation prompt to [A.9](#) while using the same prompt for answer generation. The final curated answer is again created from the sentence-level citations. Our filtering prompt [A.11](#) presents a QA pair ({QUESTION} and {ANSWER}) to the model and asks it to score the pairs' comprehensiveness, conciseness and fluency.

```

1 Task:
2 Define a single question based on the provided paragraph from the ECHR case law
  guides.
3 The paragraph must answer the question precisely and completely.
4 Think about the context of the paragraph and how it can be used to create a
  challenging legal question.
5 Use the language from the paragraph in the question.
6
7 Paragraph:
8 {PARAGRAPH}
9
10 Answer Template:
11 Thoughts: {{thoughts on the paragraph and what question could be asked}}
12 Question: {{the question that can be answered exactly by the provided paragraph}}
```

Prompt A.3.: Question Generation Single Paragraph

A. Appendix

```
1 Task:
2 Define a single question based on the provided paragraphs from the ECHR case law
  guides.
3 The paragraphs must answer the question precisely and completely.
4 Think about the context of each paragraph and what challenging legal question it
  can answer.
5 Use the language from the paragraphs in the question.
6
7 Paragraphs:
8 {PARAGRAPHS[5]}
9
10 Answer Template:
11 Thoughts: {{thoughts on the paragraphs and what question could be asked}}
12 Question: {{the question that can be answered exactly by the provided
  paragraphs}}
13 Paragraphs: {{Array of paragraph numbers needed to answer the question. Example:
  [1,2,6,7]}}
```

Prompt A.4.: Question Generation Multiple Paragraphs

```
1 Scenario:
2 Assume the role of an experienced lawyer specializing in ECHR case law.
3 Your objective is to develop educational and challenging questions for trainee
  lawyers.
4 Each question should be based on the provided paragraphs from the ECHR case law
  guides.
5 When formulating a question avoid over-contextualizing (referring to specific
  countries or asking about specific cases) and avoid asking questions about broad
  concepts for which the provided sentences only give partial answers.
6
7 Sentences:
8 {PARAGRAPH_SENTENCES[5]}
9
10 Steps:
11 1. Come up with a question that can be answered exactly with a subset of the
   provided sentences from the case law guide.
12 2. For each sentence reason about if it is required to answer the question.
13 3. Formulate an exact answer by combining the provided sentences.
14
15 Answer Template:
16 Question: {{Question that can be answered exactly with a subset of the provided
  sentences}}
17 Sentence [1]: {{Reason about why this sentence is (not) required for the answer}}
18 Sentence [2]: {{Reason about why this sentence is (not) required for the answer}}
19 ... (reason about all sentences 1 to k)
20 Sentence [k]: {{Reason about why this sentence is (not) required for the answer}}
21 Chosen Sentences: {{Array of chosen sentences. Example: [1,2,4,5,7]}}
```

Prompt A.5.: Question Generation Sentence Level Chain of Thought

A. Appendix

```
1 Case law paragraphs:
2 {PARAGRAPHS[3]}
3
4 Task:
5 Define a single challenging legal question that can be answered with the given
6 case law paragraphs.
7 Reuse the language from the case law in the question.
8
9 Question:
```

Prompt A.6.: Question Generation for Sentence-Level-CoT with Search

```
1 Case law sentences:
2 {PARAGRAPH_SENTENCES[n]}
3
4 Task: Answer the following question with the provided sentences.
5 At the end of each sentence in your answer cite the used sentences in square
6 brackets.
7
8 Question:
9 {QUESTION}
10 Answer:
```

Prompt A.7.: Answer Curation for Sentence-Level-CoT with Search

A. Appendix

```
1 Your objective is to develop educational and challenging questions for lawyers
  working with ECHR case law and for judges who want to draft judgments based on
  ECHR case law.
2 Each question should be based on the provided paragraphs from the ECHR case law
  guides.
3 When formulating a question reuse the language from the ECHR case law and match
  legal doctrines to specific facts.
4 Emphasize the patterns that link facts to specific legal doctrines.
5
6 Doctrines and facts:
7 {PARAGRAPHS[3]}
8
9 Steps:
10 1. Identify how the margin of appreciation and positive obligations apply in
    relation to the State's discretion
11 2. Identify the reasons that justify necessity and pressing social needs
12 3. Identify the reasons that command that rights be effective in their
    application
13 4. Identify how reasonable measures apply in relation to the State's discretion
    and to restrictions imposed by States or private individuals
14 5. Identify the reasons set forth by the Court to defer to domestic reasons
    provided by domestic authorities
15 6. Define a question that can be answered exactly by the given legal doctrines
    and applicable facts to those doctrines
16
17 Answer Template:
18 Margin of appreciation: {{how do the margin of appreciation and positive
    obligations apply in relation to the State's discretion}}
19 Necessities: {{reasons that justify necessity and pressing social needs}}
20 Effectivity: {{reasons that command that rights be effective in their
    application}}
21 Reasonable Measures: {{how do reasonable measures apply in relation to the
    State's discretion and to restrictions imposed by States or private
    individuals?}}
22 Domestic Reasons: {{the reasons set forth by the Court to defer to domestic
    reasons provided by domestic authorities}}
23 Question: {{define a single question that can be answered exactly by the given
    legal doctrines and applicable facts reusing the language from the ECHR case
    law}}
```

Prompt A.8.: Question Generation for Legal Sentence-Level-CoT with Search v1

A. Appendix

```
1 Your objective is to develop educational and challenging questions for lawyers
  working with ECHR case law and for judges who want to draft judgments based on
  ECHR case law.
2 Each question should be based on the provided paragraphs from the ECHR case law
  guides.
3 When formulating a question reuse the language from the ECHR case law and match
  legal doctrines to specific facts.
4 Emphasize the patterns that link facts to specific legal doctrines.
5
6 Doctrines and facts:
7 {PARAGRAPHS[3]}
8
9 Steps:
10 1. Identify what are the criteria under the Convention for applying the rights
    enshrined therein.
11 2. Identify the conditions that the Court sets forth with view to analyse the
    legality of domestic measures and restrictions
12 3. Identify the reasons provided by the Court to protect applicants and victims
    and to differentiate between them.
13 4. Identify the reasons set forth by the Court to distinguish between legal
    doctrines and contextual application of those doctrines.
14 5. Assign a set of facts to its corresponding Article and identify a sequence of
    reasons that justify the application of the Article to those facts.
15 6. Explain why analogies and comparisons between Article-fact pair are pertinent
16 7. Identify separately reasons that are linked to margin of appreciation of the
    State from those linked to the Court's appreciation of facts
17 8. Define a question that can be answered exactly by the given Article-facts
    correspondence
18
19 Answer Template:
20 Criteria for rights: {{how does the Court define the criteria for applying
    rights}}
21 Legality of domestic measures and restrictions: {{conditions that determine that
    domestic measures are compliant with the Convention}}
22 Protection and differentiation of applicants and victims: {{circumstances and
    conditions that limit or allow applicants and victims to present their case}}
23 Distinction between legal doctrines and contextual application: {{how and why
    and in what circumstances legal doctrines apply to specific facts}}
24 Article-facts correspondence: {{the reasons set forth by the Court to justify
    the application of the Article/Articles to those facts}}
25 Analogies and comparisons between Article-fact pair {{the reasons set forth by
    the Court to justify why articles and facts differ from one another}}
26 margin of appreciation of States and Court's appreciation {{the reasons,
    circumstances and conditions set forth by the Court to explain margin of
    appreciation of States and its own appreciation}}
27 Question: {{define a single question that can be answered exactly by the given
    legal doctrines and applicable facts and by the Article-fact pair, reusing the
    language from the ECHR case law and adjusting the question and the answer
    depending on how fact-Article correspondence is better addressed with What? How?
    Why?}}
```

Prompt A.9.: Question Generation for Legal Sentence-Level-CoT with Search v2

A. Appendix

```
1 Your objective is to develop educational and challenging question-answer pairs
  for lawyers working with ECHR case law and for judges who want to draft
  judgments based on ECHR case law.
2
3 Doctrines and facts:
4 {PARAGRAPH_SENTENCES[n]}
5
6 Task: Answer the following question based on the provided doctrines and facts
  from the ECHR case law guides.
7
8 Use the provided doctrines and facts to answer the question.
9 Use citations! At the end of each sentence in your answer add all the numbers of
  the used facts and doctrines in square brackets.
10
11 Question: {QUESTION}
12 Answer:
```

Prompt A.10.: Answer Curation for Legal Sentence-Level-CoT with Search

```
1 You are a strict legal expert judging ECHR legal question-answer pairs. The
  answer might be bad, so be strict!
2
3 Question: {QUESTION}
4 Potential Answer: {ANSWER}
5
6 You MUST answer each question in full sentences!
7
8 The response MUST follow this template:
9 Comprehensiveness Analysis: {{Go through the answer and analyze how well it
  answers the question. Does it cover all angles of the question? If the question
  is not a proper question or not a generic question (mentions a specific case),
  give a score of 1.}}
10 Comprehensiveness Score: {{A score from 1 (not comprehensive at all) to 5
  (extremely comprehensive)}}
11 Conciseness: {{Is there any part in the answer irrelevant/unrelated to the
  question? If so, what is unneeded?}}
12 Conciseness Score: {{A score from 1 (not concise at all) to 5 (extremely
  concise)}}
13 Answer Fluency: {{Are there any bad sentence transitions in the answer? Are the
  sentences ordered correctly? Does the answer start with text clearly continuing
  the previous text that is not there?}}
14 Answer Fluency Score: {{A score from 1 (not fluent) to 5 (perfectly fluent)}}

```

Prompt A.11.: Filtering Prompt

A.6 ECHR-QA QA Pair Curation Code

```
1 STRATEGIES = [  
2     legal_sentence_level_CoT_with_search_v2,  
3     legal_sentence_level_CoT_with_search_v1,  
4     sentence_level_CoT_with_search  
5 ]  
6  
7 echr_qa = []  
8  
9 def curate_quality_qa_pair(guide, paragraphs):  
10     # Attempts to generate a qa pair that passes the automatic evaluation  
11     for strategy in STRATEGIES:  
12         qa_pair = strategy(guide, paragraphs)  
13         if evaluate(qa_pair) >= 4:  
14             echr_qa.append(qa_pair)  
15             break  
16     # No quality QA pair could be created  
17  
18 # Loop over all the guides with a paragraph window of 3 and a stride of 2  
19 for guide in guides:  
20     for i in range(1, len(guide) - 2, steps=2):  
21         paragraphs = [i, i+1, i+2]  
22         curate_quality_qa_pair(guide, paragraphs)
```

Code A.12.: QA pair curation strategy for the ECHR-QA dataset

A.7 Attributed QA Prompts

This section presents the prompts used during our attributed QA strategies presented in Chapter 4 [Attributed QA Approaches](#).

For both of our retrieve-then-generate strategies, RAG and LLaTrieval, we used the zero-shot prompt [A.14](#). We present the retrieved `{DOCUMENTS}` preceded with enumerations `[i]` : used as separators and as identifiers for the retrieved paragraphs from our DB (see [4.1 Retrieval](#)). We heavily rely on models following our requested output format to parse the outputs across all our prompts. If the output format is not followed, we try to punish the model as little as possible; however, this is a challenging task.

In our post-hoc approaches, we used a very simple zero-shot prompt (Prompt [A.13](#)).

The LLaTrieval prompts for missing-information query (Prompt [A.15](#)), sliding document selection (Prompt [A.16](#)), and evaluation of the selected paragraphs (Prompt [A.17](#)) are only small modifications of the prompts proposed by [X. Li, Zhu, et al., 2023](#). In each, we present the LLM with retrieved `{DOCUMENTS}` and the `{QUESTION}`.

The RARR prompts: deciding if supporting evidence is required for the sentence (Prompt: [A.18](#)), the prompt to identify which retrieved documents support, contradict, or are irrelevant for the given sentence (Prompt: [A.19](#)), and the editing prompt [A.20](#) are zero-shot adaptations of the few-shot prompts proposed by [L. Gao et al., 2022](#). In addition to requiring output formats, we additionally made use of HTML tags to help the model highlight the sentence `{SENTENCE}` we are annotating in input and output. The edited sentence was required to be enclosed in the tag `<fixed-sentence-start>`. We did not experience major issues with models following our requested output formats except for SaulLM.

```
1 You are an ECHR legal expert tasked to answer the following question.
2 Question: {QUESTION}
3 Answer :
```

Prompt A.13.: Base Prompt

A. Appendix

```
1 You are an ECHR legal expert tasked to answer a question.
2 The following documents were retrieved and should help you answer the question:
3 {DOCUMENTS}
4
5 Instructions:
6 Use the retrieved documents to answer the question.
7 Reuse the language from the documents!
8 Cite relevant documents at the end of a sentence!
9 Accepted formats: sentence [citation(s)].
10 Valid citation formats: [Doc 1] or [Doc 1, Doc 2, Doc 3]
11 You must follow the [Doc i] format! Do NOT use case names or paragraph numbers
    to cite documents!
12 You should NOT provide a list of all used citations at the end of your response!
13
14 Question: {QUESTION}
15 Answer:
```

Prompt A.14.: RAG Prompt

```
1 You are a PassageRetriever,
2 capable of identifying missing content that answers the given question but does
3 not exist in the given possible answering
4 passages and then using your own knowledge to generate correct answering
5 passages using missing content you identify.
6
7 Input:
8 - Question: The specific question.
9 - Answering Passages: Possible answering passages.
10
11 Output:
12 - Correct answering passages generated using missing content you identify based
13 on your own knowledge.
14
15 Rules:
16 1. You have to use your own knowledge to generate correct answering passages
17 using missing content you identify.
18 2. Only generate the required correct answering passages. Do not output anything
19 else.
20 3. Directly use your own knowledge to generate correct answering passages if you
21 think the given possible answering passages do not answer to the given question.
22
23 Workflow:
24 1. Read and understand the question and possible answering passages.
25 2. Identify missing content that answers the given question but does not exist
26 in the given possible answering passages.
27 3. Directly use your own knowledge to generate correct answering passages if you
28 think the given possible answering passages do not answer to the given
29 question(s).
30 4. Use your own knowledge to generate correct answering passages using missing
31 content you identify.
32
33 Answering Passages:
34 {DOCUMENTS}
35
36 Question:
37 {QUESTION}
38
39 Output Format: (You MUST follow this output format!)
40 Correct Answering Passages: [correct answering passages]
41 Missing Passages: [missing passages]
```

Prompt A.15.: Prompt to retrieve new passages: a modified version of the passage retrieval prompt proposed by [X. Li, Zhu, et al., 2023](#) for LLaTrieval

A. Appendix

```
1 You are DocSelectorGPT, capable of selecting a specified number of documents for
  answering the user's specific question.
2
3 Input:
4 - Question: The specific question
5 - Candidate Documents: Documents contain supporting documents which can support
  answering the given questions. Candidate documents will have their own
  identifiers for FactRetrieverGPT to cite.
6
7 Skill:
8 1. Analyzing the given question and understanding the required information.
9 2. Searching through candidate documents to select k supporting documents whose
  combination can maximally support giving a direct, accurate, clear and engaging
  answer to the question and make the answer and is closely related to the core of
  the question.
10
11 Workflow:
12 1. Read and understand the questions posed by the user.
13 2. Browse through candidate documents to select k documents whose combination
  can maximally support giving a direct, accurate, clear and engaging answer to
  the question(s) and make the answer and is closely related to the core of the
  question.
14 3. List all selected documents.
15
16 Output:
17 - Selected Documents: The identifiers of selected supporting documents whose
  combination can maximally support giving an accurate and engaging answer to the
  question and make the answer and is closely related to the core of the question.
18
19 Output Example:
20 Selected Documents: Doc 2, Doc 6, Doc 8 (You MUST follow this format!)
21
22 Max number of selectable documents: {k}
23 - You can only select a maximum of {k} documents!
24
25 Candidate Documents:
26 {DOCUMENTS}
27
28 Question:
29 {QUESTION}
30
31 Output Format (You MUST follow this output format!)
32 Thoughts: [Your thoughts about which candidate documents support the question
  well and why]
33 Selected Documents: [document identifiers]
```

Prompt A.16.: Prompt to select documents: a modified version of the selection prompt proposed by [X. Li, Zhu, et al., 2023](#) for LLaTrieval

```
1 You are ScoreGPT as introduced below.
2 You are ScoreGPT, capable of scoring candidate documents based on their level of
  support for the corresponding question, with a rating range from 0 to 10.
3
4 Input:
5 - Question: The specific question.
6 - Candidate Documents: Documents whose combination may maximally support the
  corresponding question.
7
8 Skill:
9 1. Analyzing the given question(s) and understanding the required information.
10 2. Searching through documents to score them based on their level of support for
   the corresponding question(s),
11 with a rating range from 0 to 10.
12
13 Output:
14 - A score ranging from 0 to 10, where a higher score indicates greater support
   of the candidate documents for the corresponding question, and a lower score
   indicates lesser support.
15
16 Candidate Documents:
17 {DOCUMENTS}
18
19 Question:
20 {QUESTION}
21
22 Output Format: (You MUST follow this output format!)
23 Thoughts: [Your thoughts about how well the candidate documents support the
   question]
24 Score: [SCORE]
```

Prompt A.17.: Prompt to score documents: a modified version of the scoring prompt proposed by [X. Li, Zhu, et al., 2023](#) for LLaTrieval

A. Appendix

```
1 You will be determining if a sentence of the answer should be supported by a
  case law citation.
2
3 ## Context
4 Question: {QUESTION}
5 Answer: {ANSWER}
6
7 ## Now we will analyze if the following sentence should have a citation
8
9 The sentence for which to decide if it should have a citation:
10 <sentence-start>{SENTENCE}</sentence-end>
11
12 First, analyze the sentence and decide if it should be supported by a case law
  citation.
13 Note:
14 - General knowledge, headers, and other non-sentences do not require citations.
15 - Legal arguments, facts, examples, ... should be supported by citations.
16
17 The format of your response MUST look like this:
18 Thoughts: [Reason why the sentence <sentence-start>{SENTENCE}</sentence-end>
  should have a citation]
19 Should have a supporting citation: [Yes/No]
```

Prompt A.18.: Sentence needs evidence prompt of our RARR ([L. Gao et al., 2022](#)) inspired approach

A. Appendix

```
1 You will be determining if a piece of evidence agrees with, disagrees with, or
  is irrelevant to a sentence for a given question.
2
3 ## Context
4 Question: {QUESTION}
5 Answer: {ANSWER}
6
7 ## Now we will analyze the evidence for the following sentence in the answer
8
9 Evidence:
10 <evidence-start>{DOCUMENT}</evidence-end>
11
12 The sentence for which to decide if the evidence agrees, disagrees, or is
  irrelevant:
13 <sentence-start>{SENTENCE}</sentence-end>
14
15 Carefully analyze the evidence and explain in a reasoning step whether it
  agrees, contradicts, or is irrelevant to the sentence in
  <sentence-start>...</sentence-end> tags.
16 Then, based on your reasoning, provide your final classification, which MUST be
  one of the following:
17 - Agrees
18 - Disagrees
19 - Irrelevant
20
21 The format of your response MUST look like this:
22 Thoughts: [Reason whether the evidence agrees, disagrees, or is irrelevant to
  the given sentence]
23 Final classification: [Your final classification here:
  Agrees/Disagrees/Irrelevant]
```

Prompt A.19.: Evidence relevancy prompt of our RARR ([L. Gao et al., 2022](#)) inspired approach

```
1 You will be editing a sentence based on the disagreement with the evidence.
2
3 ## Context
4 Question: {QUESTION}
5 Answer: {ANSWER}
6
7 ## Now we will edit the following sentence in the answer based on the
disagreement with the evidence
8
9 Evidence:
10 <evidence-start>{DOCUMENT}</evidence-end>
11
12 The sentence in the answer the evidence disagrees with:
13 <sentence-start>{SENTENCE}</sentence-end>
14
15 First, carefully analyze the sentence and identify the part that contains the
disagreement with the evidence.
16 Then, rewrite the sentence with MINIMAL modification to resolve the disagreement.
17 We will not accept drastic changes to the sentence!
18
19 The format of your response MUST look like this:
20 Thoughts: [Reason why the sentence in <sentence-start>...</sentence-end> should
be edited]
21 Fix with minimal edit: [The corrected entire sentence with MINIMAL modification
to resolve the disagreement enclosed by
<fixed-sentence-start>...<fixed-sentence-end>]
```

Prompt A.20.: Editing prompt of our RARR ([L. Gao et al., 2022](#)) inspired approach

A.8 Annotation Platform

ECHR-QA Annotation Platform
⚙️ Log Out

Target Answer

Question:

Under what circumstances can individual members of a public body bring defamation proceedings in their own individual name, according to the principles set forth by the Court in relation to the protection of reputation and freedom of expression?

Target Answer:

This does not exclude, however, that individual members of a public body, who could be "easily identifiable" in view of the limited number of its members and the nature of the allegations made against them, may be entitled to bring defamation proceedings in their own individual name. Civil servants in order for Article 8 to come into play, however, an attack on a person's reputation must attain a certain level of seriousness and in a manner causing prejudice to personal enjoyment of the right to respect for private life (Bédat v. Switzerland [GC], § 72; Axel Springer AG v. Germany [GC], § 83; A. v. Norway, § 64). The Court considers that the "limits of acceptable criticism" are much wider as regards individuals with a public status than as regards private individuals (Palomo Sánchez and Others v. Spain [GC], § 71). The Court considers that State bodies and civil servants acting in an official capacity have to accept that they are subject to wider limits of acceptable criticism than private individuals (Romanenko and Others v. Russia, § 47; Toranzo Gomez v. Spain, § 65; see also Frisk and Jensen v. Denmark, § 56, concerning criticism of a public hospital, and Lombardo and Others v. Malta, § 54, a local council). The same principles apply to institutions responsible for providing a public service.

Citations:

[BÉDAT V. SWITZERLAND](#)

§ 72

The Court reiterates that the right to protection of reputation is a right which is protected by Article 8 of the Convention as part of the right to respect for private life (see Chauvy and Others v. France, no. 64915/01, § 70, ECHR 2004VI; Polanco Torres and Movilla Polanco v. Spain, no. 34147/06 § 40, 21 September 2010; and Axel Springer AG, cited above, § 83). The concept of "private life" is a broad term which is not susceptible to exhaustive definition. It covers the physical and psychological integrity of a person, and can therefore embrace multiple aspects of the person's identity such as, for example, ...

1/5

Annotation Form

Correctness
Rank the experiments based on the accuracy, completeness, and relevance of their answers to the given question.

A: B: C: D:

Groundedness
Rank the experiments by how constitutively their answers are supported by citations.

A: B: C: D:

Source Relevance
Rank the experiments by how relevant and directly supportive the cited sources are in answering the question.

A: B: C: D:

Better than Target
Check any generated answers that are better than the target answer

☐ A ☐ B ☒ C ☐ D

Note ⓘ

Same considerations as for 5. A and D equivalent for undergrad B and C more targeted with supportive examples. C is better in my opinion for the way in which it describes and justifies the conditions and links with other possible interests (burden of media) and that they are not under 10.

[Submit](#)

< Previous
... 5 6 7 ...
Next >

Figure A.21.: ECHR-QA Expert Annotation Platform (Experiment Answer)

ECHR-QA Annotation Platform

Log Out

Experiment B

Generated Answer:

Individual members of a public body may be entitled to bring defamation proceedings in their own individual name if they are "easily identifiable" and the allegations made against them are of a nature that would warrant such action (CASE OF OOO MEMO v. RUSSIA § 47). The Court has examined the proportionality of interferences with the right to freedom of expression in defamation cases, considering elements such as the nature of the statements, the context in which they were made, the extent to which they affected the individuals concerned, and the severity of the sanctions imposed (CASE OF SAPUNDZHIEV v. BULGARIA § 39). When assessing the legitimacy of the aim to protect the reputation or rights of others, the existence of an objective link between the impugned statement and the person suing in defamation is a requisite element (CASE OF FILATENKO v. RUSSIA § 45). However, public bodies and authorities, such as courts' management departments, have been found to fall outside the category of 'others' in Article 10 § 2 of the Convention (CASE OF ROMANENKO AND OTHERS v. RUSSIA § 39). Policymakers in various jurisdictions have argued that public bodies should not have standing to sue in defamation, and the Court has acknowledged the existence of sound policy reasons for this position (CASE OF ROMANENKO AND OTHERS v. RUSSIA § 39). These principles apply to both television and radio broadcasting (CASE OF OOO MEMO v. RUSSIA § 41).

Citations:

[CASE OF OOO MEMO V. RUSSIA](#)

§ 47

. It follows that civil defamation proceedings brought, in its own name, by a legal entity that exercises public power may not, as a general rule, be regarded as being in pursuance of the legitimate aim of "the protection of the reputation ... of others" under Article 10 § 2 of the Convention. This does not rule out the possibility that individual members of a public body who could be "easily identifiable" in view of the limited number of its members and the nature of the allegations made against them (see Thoma v. Luxembourg , no. 38432/97, § 56, ECHR 2001 - III, and, mutatis mutandis , Lombardo and Others , cited above, § 54) may be entitled to bring defamation proceedings in their own name.

[CASE OF SAPUNDZHIEV V. BULGARIA](#)

§ 39

In cases where individuals have been found guilty of defamation for complaints they had

3/5

Annotation Form

Correctness

Rank the experiments based on the accuracy, completeness, and relevance of their answers to the given question.

A: 4 B: 2 C: 1 D: 3

Groundedness

Rank the experiments by how constitutively their answers are supported by citations.

A: 4 B: 3 C: 1 D: 2

Source Relevance

Rank the experiments by how relevant and directly supportive the cited sources are in answering the question.

A: 4 B: 3 C: 1 D: 2

Better than Target

Check any generated answers that are better than the target answer

☐ A ☐ B ☒ C ☐ D

Note

Same considerations as for 5. A and D equivalent for undergrad B and C more targeted with supportive examples. C is better in my opinion for the way in which it describes and justifies the conditions and links with other possible interests (burden of media) and that they are not under 10.

Submit

< Previous

...

5

6

7

...

Next >

Figure A.22.: ECHR-QA Expert Annotation Platform (Experiment Answer)

List of Figures

3.1 (Legal) CoT with Search	10
A.1 ECHR-QA Data Analysis	51
A.2 Expert QA Pair Evaluation Distributions	52
A.3 Question Generation Single Paragraph	53
A.4 Question Generation Multiple Paragraphs	54
A.5 Question Generation Sentence Level Chain of Thought	54
A.6 Question Generation for Sentence-Level-CoT with Search	55
A.7 Answer Curation for Sentence-Level-CoT with Search	55
A.8 Question Generation for Legal Sentence-Level-CoT with Search v1	56
A.9 Question Generation for Legal Sentence-Level-CoT with Search v2	57
A.10 Answer Curation for Legal Sentence-Level-CoT with Search	58
A.11 Filtering Prompt	58
A.12 QA pair curation strategy for the ECHR-QA dataset	59
A.13 Base Prompt	60
A.14 RAG Prompt	61
A.15 Prompt to retrieve new passages: a modified version of the passage retrieval prompt proposed by X. Li, Zhu, et al., 2023 for LLattribution	62
A.16 Prompt to select documents: a modified version of the selection prompt proposed by X. Li, Zhu, et al., 2023 for LLattribution	63
A.17 Prompt to score documents: a modified version of the scoring prompt proposed by X. Li, Zhu, et al., 2023 for LLattribution	64
A.18 Sentence needs evidence prompt of our RARR (L. Gao et al., 2022) inspired approach	65
A.19 Evidence relevancy prompt of our RARR (L. Gao et al., 2022) inspired approach	66
A.20 Editing prompt of our RARR (L. Gao et al., 2022) inspired approach	67
A.21 ECHR-QA Expert Annotation Platform (Experiment Answer)	68
A.22 ECHR-QA Expert Annotation Platform (Experiment Answer)	69

List of Tables

3.1	Expert Evaluation Scores	11
3.2	Filtering Results	13
5.1	Comparison of different LLMs across attributed QA strategies on ECHR-QA . .	21
5.2	Edits by RARR	22
5.3	Comparison of avg. answer tokens, sentences, and citations of different LLMs across attributed QA strategies	23
5.4	Average expert ranking compared to auto ranking for Mistral-7B across methods	24
5.5	Average expert ranking compared to auto ranking for RAG across models . . .	24
5.6	Correlation and p-value of expert rankings compared to auto rankings	25
5.7	Percentage of answers the expert labeled as better than the target answer from the 160 generated answers in the 40 annotations.	26
A.1	QA pair 27 from ECHR-QA	38
A.2	QA pair 43 from ECHR-QA	39
A.3	QA pair 60 from ECHR-QA	39
A.4	QA pair 62 from ECHR-QA	40
A.5	QA pair 76 from ECHR-QA	40
A.6	QA pair 77 from ECHR-QA	40
A.7	QA pair 119 from ECHR-QA	41
A.8	QA pair 122 from ECHR-QA	41
A.9	QA pair 123 from ECHR-QA	42
A.10	Example expert annotated QA pair generated with Legal Sentence-Level-CoT with Search v1 for Guide Art 3	42
A.11	Example Mistral Answers	46
A.12	Example RAG Answers	47
A.13	Example Edit by Saul using Post-hoc Retrieval - RARR	48
A.14	Example Edit by Llama 8b using Post-hoc Retrieval - RARR	48
A.15	Example Edit by Llama 8b using Post-hoc Retrieval - RARR	49
A.16	GTR vs. BM25 for RAG Llama-3-70B	50

Acronyms

ECHR European Convention on Human Rights.

ECtHR European Court of Human Rights.

IRAC Issue-Rule-Application-Conclusion.

LLM Large Language Model.

NLI Natural Language Inference.

QA Question Answering.

Bibliography

Abdallah, A., Piryani, B., & Jatowt, A. (2023). Exploring the state of the art in legal qa systems. *Journal of Big Data*, 10(1), 127.

Ahmad, W. U., Chi, J., Tian, Y., & Chang, K.-W. (2020). Policyqa: A reading comprehension dataset for privacy policies. *arXiv preprint arXiv:2010.02557*.

Aletras, N., Tsarapatsanis, D., Preotjiuc-Pietro, D., & Lampos, V. (2016). Predicting judicial decisions of the european court of human rights: A natural language processing perspective. *PeerJ computer science*, 2, e93.

Amouyal, S. J., Wolfson, T., Rubin, O., Yoran, O., Herzig, J., & Berant, J. (2022). Qampari: An open-domain question answering benchmark for questions with many answers from multiple paragraphs. *arXiv preprint arXiv:2205.12665*.

Bach, N. X., Thien, T. H. N., Phuong, T. M., et al. (2017). Question analysis for vietnamese legal question answering. *2017 9th International Conference on Knowledge and Systems Engineering (KSE)*, 154–159.

Blair-Stanek, A., Holzenberger, N., & Van Durme, B. (2023). Can gpt-3 perform statutory reasoning? *Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law*, 22–31.

Bohnet, B., Tran, V. Q., Verga, P., Aharoni, R., Andor, D., Soares, L. B., Ciaramita, M., Eisenstein, J., Ganchev, K., Herzig, J., et al. (2022). Attributed question answering: Evaluation and modeling for attributed large language models. *arXiv preprint arXiv:2212.08037*.

Brown, T. B. (2020). Language models are few-shot learners. *arXiv preprint ArXiv:2005.14165*.

Büttner, M., & Habernal, I. (2024). Answering legal questions from laymen in german civil law system. *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2015–2027.

Chalkidis, I., Androutsopoulos, I., & Aletras, N. (2019). Neural legal judgment prediction in english. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 4317–4323.

Chalkidis, I., Fergadiotis, M., Tsarapatsanis, D., Aletras, N., Androutsopoulos, I., & Malakasiotis, P. (2021). Paragraph-level rationale extraction through regularization: A case study on european court of human rights cases. *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 226–241.

Chen, A., Yao, F., Zhao, X., Zhang, Y., Sun, C., Liu, Y., & Shen, W. (2023). Equals: A real-world dataset for legal question answering via reading chinese laws. *Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law*, 71–80.

Cheong, I., Xia, K., Feng, K. K., Chen, Q. Z., & Zhang, A. X. (2024). (a) i am not a lawyer, but...: Engaging legal experts towards responsible llm policies for legal advice. *The 2024 ACM Conference on Fairness, Accountability, and Transparency*, 2454–2469.

Chern, S., Chern, E., Neubig, G., & Liu, P. (2024). Can large language models be trusted for evaluation? scalable meta-evaluation of llms as evaluators via agent debate. *arXiv preprint arXiv:2401.16788*.

Choi, J. H., & Schwarcz, D. (2023). Ai assistance in legal analysis: An empirical study. *Available at SSRN 4539836*.

Colombo, P., Pires, T. P., Boudiaf, M., Culver, D., Melo, R., Corro, C., Martins, A. F., Esposito, F., Raposo, V. L., Morgado, S., et al. (2024). Saullm-7b: A pioneering large language model for law. *arXiv preprint arXiv:2403.03883*.

Dahl, M., Magesh, V., Suzgun, M., & Ho, D. E. (2024). Large legal fictions: Profiling legal hallucinations in large language models journal of legal analysis (forthcoming).

Duan, X., Wang, B., Wang, Z., Ma, W., Cui, Y., Wu, D., Wang, S., Liu, T., Huo, T., Hu, Z., et al. (2019). Cjrc: A reliable human-annotated benchmark dataset for chinese judicial reading comprehension. *Chinese Computational Linguistics: 18th China National Conference, CCL 2019, Kunming, China, October 18–20, 2019, Proceedings 18*, 439–451.

Fagan, F. (2024). A view of how language models will transform law. *arXiv preprint arXiv:2405.07826*.

Fan, A., Jernite, Y., Perez, E., Grangier, D., Weston, J., & Auli, M. (2019). Eli5: Long form question answering. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 3558–3567.

Fawei, B., Pan, J. Z., Kollingbaum, M., & Wyner, A. Z. (2019). A semi-automated ontology construction for legal question answering. *New Generation Computing*, 37(4), 453–478.

Filtz, E., Navas-Loro, M., Santos, C., Polleres, A., & Kirrane, S. (2020). Events matter: Extraction of events from court decisions. *Legal Knowledge and Information Systems*, 33–42.

- Frankenreiter, J., & Nyarko, J. (2022). Natural language processing in legal tech. *Legal Tech and the Future of Civil Justice* (David Engstrom ed.) Forthcoming.
- Fu, J., Ng, S.-K., Jiang, Z., & Liu, P. (2023). Gptscore: Evaluate as you desire. *arXiv preprint arXiv:2302.04166*.
- Funkquist, M., Kuznetsov, I., Hou, Y., & Gurevych, I. (2022). Citebench: A benchmark for scientific citation text generation. *arXiv preprint arXiv:2212.09577*.
- Gallegos, I. O., Rossi, R. A., Barrow, J., Tanjim, M. M., Kim, S., Dernoncourt, F., Yu, T., Zhang, R., & Ahmed, N. K. (2024). Bias and fairness in large language models: A survey. *Computational Linguistics*, 1–79.
- Gao, L., Dai, Z., Pasupat, P., Chen, A., Chaganty, A. T., Fan, Y., Zhao, V. Y., Lao, N., Lee, H., Juan, D.-C., et al. (2022). Rarr: Researching and revising what language models say, using language models. *arXiv preprint arXiv:2210.08726*.
- Gao, T., Yen, H., Yu, J., & Chen, D. (2023). Enabling large language models to generate text with citations. *arXiv preprint arXiv:2305.14627*.
- Gekhman, Z., Herzig, J., Aharoni, R., Elkind, C., & Szpektor, I. (2023). Trueteacher: Learning factual consistency evaluation with large language models.
- Gramopadhye, O., Nachane, S. S., Chanda, P., Ramakrishnan, G., Jadhav, K. S., Nandwani, Y., Raghu, D., & Joshi, S. (2024). Few shot chain-of-thought driven reasoning to prompt llms for open ended medical question answering. *arXiv preprint arXiv:2403.04890*.
- Guha, N., Nyarko, J., Ho, D., Ré, C., Chilton, A., Chohlas-Wood, A., Peters, A., Waldon, B., Rockmore, D., Zambrano, D., et al. (2024). Legalbench: A collaboratively built benchmark for measuring legal reasoning in large language models. *Advances in Neural Information Processing Systems*, 36.
- Habernal, I., Faber, D., Recchia, N., Bretthauer, S., Gurevych, I., Spiecker genannt Döhmann, I., & Burchard, C. (2023). Mining legal arguments in court decisions. *Artificial Intelligence and Law*, 1–38.
- Hanna, M., & Bojar, O. (2021). A fine-grained analysis of bertscore. *Proceedings of the Sixth Conference on Machine Translation*, 507–517.
- He, H., Zhang, H., & Roth, D. (2022). Rethinking with retrieval: Faithful large language model inference. *arXiv preprint arXiv:2301.00303*.
- Hendrycks, D., Burns, C., Chen, A., & Ball, S. (2021). Cuad: An expert-annotated nlp dataset for legal contract review. *arXiv preprint arXiv:2103.06268*.

- Hennes, E. P., & Dang, L. (2021). The devil we know: Legal precedent and the preservation of injustice. *Policy insights from the behavioral and brain sciences*, 8(1), 76–83.
- Honovich, O., Aharoni, R., Herzig, J., Taitelbaum, H., Kukliansy, D., Cohen, V., Scialom, T., Szpektor, I., Hassidim, A., & Matias, Y. (2022). True: Re-evaluating factual consistency evaluation. *arXiv preprint arXiv:2204.04991*.
- Hoppe, C., Pelkmann, D., Migenda, N., Hötte, D., & Schenck, W. (2021). Towards intelligent legal advisors for document retrieval and question-answering in german legal documents. *2021 IEEE Fourth International Conference on Artificial Intelligence and Knowledge Engineering (AIKE)*, 29–32.
- Huang, W., Jiang, J., Qu, Q., & Yang, M. (2020). Aila: A question answering system in the legal domain. *IJCAI*, 5258–5260.
- Huo, S., Arabzadeh, N., & Clarke, C. (2023). Retrieving supporting evidence for generative question answering. *Proceedings of the Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*, 11–20.
- Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., Casas, D. d. I., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., et al. (2023). Mistral 7b. *arXiv preprint arXiv:2310.06825*.
- Joshi, M., Choi, E., Weld, D., & Zettlemoyer, L. (2017). TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In R. Barzilay & M.-Y. Kan (Eds.), *Proceedings of the 55th annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 1601–1611). Association for Computational Linguistics.
- Kamalloo, E., Jafari, A., Zhang, X., Thakur, N., & Lin, J. (2023). Hagrid: A human-llm collaborative dataset for generative information-seeking with attribution. *arXiv preprint arXiv:2307.16883*.
- Katz, D. M., Bommarito, M. J., Gao, S., & Arredondo, P. (2024). Gpt-4 passes the bar exam. *Philosophical Transactions of the Royal Society A*, 382(2270), 20230254.
- Kien, P. M., Nguyen, H.-T., Bach, N. X., Tran, V., Le Nguyen, M., & Phuong, T. M. (2020). Answering legal questions by learning neural attentive text representation. *Proceedings of the 28th International Conference on Computational Linguistics*, 988–998.
- Kim, M.-Y., Goebel, R., & Ken, S. (2015). Collee-2015: Evaluation of legal question answering. *Ninth International Workshop on Juris-informatics (JURISIN 2015)*.
- Köksal, A., Schick, T., Korhonen, A., & Schuetze, H. (2024). Longform: Effective instruction tuning with reverse instructions. *ICLR 2024 Workshop on Navigating and Addressing Data Problems for Foundation Models*.

- Krishna, K., Roy, A., & Iyyer, M. (2021). Hurdles to progress in long-form question answering. In K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tur, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, & Y. Zhou (Eds.), *Proceedings of the 2021 conference of the north american chapter of the association for computational linguistics: Human language technologies* (pp. 4940–4957). Association for Computational Linguistics.
- Kwiatkowski, T., Palomaki, J., Redfield, O., Collins, M., Parikh, A., Alberti, C., Epstein, D., Polosukhin, I., Devlin, J., Lee, K., et al. (2019). Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7, 453–466.
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., et al. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33, 9459–9474.
- Li, X., Zhu, C., Li, L., Yin, Z., Sun, T., & Qiu, X. (2023). Llatrival: Llm-verified retrieval for verifiable generation. *arXiv preprint arXiv:2311.07838*.
- Li, X., Cao, Y., Pan, L., Ma, Y., & Sun, A. (2023). Towards verifiable generation: A benchmark for knowledge-aware language model attribution. *arXiv preprint arXiv:2310.05634*.
- Li, X., Zhang, T., Dubois, Y., Taori, R., Gulrajani, I., Guestrin, C., Liang, P., & Hashimoto, T. B. (2023). AlpacaEval: An automatic evaluator of instruction-following models.
- Li, Y., Yue, X., Liao, Z., & Sun, H. (2024). Attributionbench: How hard is automatic attribution evaluation? *arXiv preprint arXiv:2402.15089*.
- Liang, Y., Wang, J., Zhu, H., Wang, L., Qian, W., & Lan, Y. (2023). Prompting large language models with chain-of-thought for few-shot knowledge base question generation. *arXiv preprint arXiv:2310.08395*.
- Lin, C.-Y. (2004). Rouge: A package for automatic evaluation of summaries. *Text summarization branches out*, 74–81.
- Liu, J., Ke, W., Wang, P., Shang, Z., Gao, J., Li, G., Ji, K., & Liu, Y. (2024). Towards continual knowledge graph embedding via incremental distillation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(8), 8759–8768.
- Liu, N. F., Zhang, T., & Liang, P. (2023). Evaluating verifiability in generative search engines. *arXiv preprint arXiv:2304.09848*.
- Liu, Y., Iyer, D., Xu, Y., Wang, S., Xu, R., & Zhu, C. (2023). G-eval: Nlg evaluation using gpt-4 with better human alignment. *The 2023 Conference on Empirical Methods in Natural Language Processing*.

- Livermore, M. A., Herron, F., & Rockmore, D. (2023). Language model interpretability and empirical legal studies. *Virginia Public Law and Legal Theory Research Paper*, (2023-69).
- Louis, A., & Spanakis, G. (2021). A statutory article retrieval dataset in french. *arXiv preprint arXiv:2108.11792*.
- Louis, A., van Dijck, G., & Spanakis, G. (2024). Interpretable long-form legal question answering with retrieval-augmented large language models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(20), 22266–22275.
- Magesh, V., Surani, F., Dahl, M., Suzgun, M., Manning, C. D., & Ho, D. E. (2024). Hallucination-free? assessing the reliability of leading ai legal research tools. *arXiv preprint arXiv:2405.20362*.
- Malaviya, C., Lee, S., Chen, S., Sieber, E., Yatskar, M., & Roth, D. (2023). Expertqa: Expert-curated questions and attributed answers. *arXiv preprint arXiv:2309.07852*.
- Martin, L., Whitehouse, N., Yiu, S., Catterson, L., & Perera, R. (2024). Better call gpt, comparing large language models against lawyers. *arXiv preprint arXiv:2401.16212*.
- Mochales, R., & Moens, M.-F. (2008). Study on the structure of argumentation in case law. *Proceedings of the 2008 conference on legal knowledge and information systems*, 11–20.
- Morimoto, A., Kubo, D., Sato, M., Shindo, H., & Matsumoto, Y. (2017). Legal question answering system using neural attention. *COLIEE@ ICAIL, 2017*, 79–89.
- Motoki, F., Pinho Neto, V., & Rodrigues, V. (2024). More human than human: Measuring chatgpt political bias. *Public Choice*, 198(1), 3–23.
- Mukaka, M. M. (2012). A guide to appropriate use of correlation coefficient in medical research. *Malawi medical journal*, 24(3), 69–71.
- Muller, B., Wieting, J., Clark, J. H., Kwiatkowski, T., Ruder, S., Soares, L. B., Aharoni, R., Herzig, J., & Wang, X. (2023). Evaluating and modeling attribution for cross-lingual question answering. *arXiv preprint arXiv:2305.14332*.
- Nakano, R., Hilton, J., Balaji, S., Wu, J., Ouyang, L., Kim, C., Hesse, C., Jain, S., Kosaraju, V., Saunders, W., et al. (2021). Webgpt: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332*.
- Navas-Loro, M., & Rodriguez-Doncel, V. (2022). Whenthefact: Extracting events from european legal decisions. In *Legal knowledge and information systems* (pp. 219–224). IOS Press.
- Ni, J., Qu, C., Lu, J., Dai, Z., Abrego, G. H., Ma, J., Zhao, V., Luan, Y., Hall, K., Chang, M.-W., et al. (2022). Large dual encoders are generalizable retrievers. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 9844–9855.

- Pan, Q., Ashktorab, Z., Desmond, M., Cooper, M. S., Johnson, J., Nair, R., Daly, E., & Geyer, W. (2024). Human-centered design recommendations for llm-as-a-judge. *arXiv preprint arXiv:2407.03479*.
- Pillutla, K., Swayamdipta, S., Zellers, R., Thickstun, J., Welleck, S., Choi, Y., & Harchaoui, Z. (2021). Mauve: Measuring the gap between neural text and human text using divergence frontiers. *Advances in Neural Information Processing Systems*, 34, 4816–4828.
- Poudyal, P., Gonçalves, T., & Quaresma, P. (2019). Using clustering techniques to identify arguments in legal documents. *ASAIL@ ICAIL*, 2385.
- Poudyal, P., Šavelka, J., Ieven, A., Moens, M. F., Gonçalves, T., & Quaresma, P. (2020). Echr: Legal corpus for argument mining. *Proceedings of the 7th Workshop on Argument Mining*, 67–75.
- Qin, Y., Cai, Z., Jin, D., Yan, L., Liang, S., Zhu, K., Lin, Y., Han, X., Ding, N., Wang, H., et al. (2023). Webcpm: Interactive web search for chinese long-form question answering. *arXiv preprint arXiv:2305.06849*.
- Quemy, A., & Wrembel, R. (2021). Echr-od: On building an integrated open repository of legal documents for machine learning applications. *Information Systems*, 101822.
- Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., & Liu, P. J. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140), 1–67.
- Rajpurkar, P., Jia, R., & Liang, P. (2018). Know what you don't know: Unanswerable questions for squad. *arXiv preprint arXiv:1806.03822*.
- Rashkin, H., Nikolaev, V., Lamm, M., Aroyo, L., Collins, M., Das, D., Petrov, S., Tomar, G. S., Turc, I., & Reitter, D. (2023). Measuring attribution in natural language generation models. *Computational Linguistics*, 49(4), 777–840.
- Ravichander, A., Black, A. W., Wilson, S., Norton, T., & Sadeh, N. (2019). Question answering for privacy policies: Combining computational and legal perspectives. *arXiv preprint arXiv:1911.00841*.
- Rodgers, I., Armour, J., & Sako, M. (2023). How technology is (or is not) transforming law firms. *Annual Review of Law and Social Science*, 19(1), 299–317.
- Santosh, T., Aly, M., & Grabmair, M. (2024). Lexabsumm: Aspect-based summarization of legal decisions. *arXiv preprint arXiv:2404.00594*.

- Santosh, T., Haddad, R. G., & Grabmair, M. (2024). Ecthr-pcr: A dataset for precedent understanding and prior case retrieval in the european court of human rights. *arXiv preprint arXiv:2404.00596*.
- Santosh, T., Hernandez, E. Q., & Grabmair, M. (2024). Query-driven relevant paragraph extraction from legal judgments. *arXiv preprint arXiv:2404.00595*.
- Santosh, T., Ichim, O., & Grabmair, M. (2023). Zero-shot transfer of article-aware legal outcome classification for european court of human rights cases. *Findings of the Association for Computational Linguistics: EACL 2023*, 593–605.
- Santosh, T., San Blas, M. P., Kemper, P., & Grabmair, M. (2023). Leveraging task dependency and contrastive learning for case outcome classification on european court of human rights cases. *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, 1103–1111.
- Santosh, T., Xu, S., Ichim, O., & Grabmair, M. (2022). Deconfounding legal judgment prediction for european court of human rights cases towards better alignment with experts. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 1120–1138.
- Schulhoff, S., Ilie, M., Balepur, N., Kahadze, K., Liu, A., Si, C., Li, Y., Gupta, A., Han, H., Schulhoff, S., et al. (2024). The prompt report: A systematic survey of prompting techniques. *arXiv preprint arXiv:2406.06608*.
- Shuster, K., Poff, S., Chen, M., Kiela, D., & Weston, J. (2021). Retrieval augmentation reduces hallucination in conversation. *arXiv preprint arXiv:2104.07567*.
- Sovrano, F., Palmirani, M., Distefano, B., Sapienza, S., & Vitali, F. (2021). A dataset for evaluating legal question answering on private international law. *Proceedings of the Eighteenth International Conference on Artificial Intelligence and Law*, 230–234.
- Spearman, C. (1961). The proof and measurement of association between two things.
- Stelmakh, I., Luan, Y., Dhingra, B., & Chang, M.-W. (2022). Asqa: Factoid questions meet long-form answers. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 8273–8288.
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. (2023). Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.
- Trivedi, P. (2018). A question answering system on regulatory documents. *Legal Knowledge and Information Systems: JURIX 2018: The Thirty-first Annual Conference*, 313, 41.

- Ushio, A., Alva-Manchego, F., & Camacho-Collados, J. (2023). An empirical comparison of lm-based question and answer generation methods. *arXiv preprint arXiv:2305.17002*.
- Valvoda, J., Cotterell, R., & Teufel, S. (2023). On the role of negative precedent in legal outcome prediction. *Transactions of the Association for Computational Linguistics*, 11, 34–48.
- Wang, J., Liang, Y., Meng, F., Sun, Z., Shi, H., Li, Z., Xu, J., Qu, J., & Zhou, J. (2023). Is chatgpt a good nlg evaluator? a preliminary study. *arXiv preprint arXiv:2303.04048*.
- Wang, S. H., Scardigli, A., Tang, L., Chen, W., Levkin, D., Chen, A., Ball, S., Woodside, T., Zhang, O., & Hendrycks, D. (2023). Maud: An expert-annotated legal nlp dataset for merger agreement understanding. *arXiv preprint arXiv:2301.00876*.
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., Zhou, D., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35, 24824–24837.
- Xu, S., Ichim, O., Risini, I., Plank, B., Grabmair, M., et al. (2023). From dissonance to insights: Dissecting disagreements in rationale construction for case outcome classification. *arXiv preprint arXiv:2310.11878*.
- Xu, S., Staufer, L., Ichim, O., Heri, C., Grabmair, M., et al. (2023). Vechr: A dataset for explainable and robust classification of vulnerability type in the european court of human rights. *arXiv preprint arXiv:2310.11368*.
- Xu, Z., Jain, S., & Kankanhalli, M. (2024). Hallucination is inevitable: An innate limitation of large language models. *arXiv preprint arXiv:2401.11817*.
- Yang, Z., Qi, P., Zhang, S., Bengio, Y., Cohen, W. W., Salakhutdinov, R., & Manning, C. D. (2018). Hotpotqa: A dataset for diverse, explainable multi-hop question answering. *arXiv preprint arXiv:1809.09600*.
- Yue, X., Wang, B., Zhang, K., Chen, Z., Su, Y., & Sun, H. (2023). Automatic evaluation of attribution by large language models. *arXiv preprint arXiv:2305.06311*.
- Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., & Artzi, Y. (2019). Bertscore: Evaluating text generation with bert. *arXiv preprint arXiv:1904.09675*.
- Zhang, X., Thakur, N., Ogundepo, O., Kamalloo, E., Alfonso-Hermelo, D., Li, X., Liu, Q., Rezagholizadeh, M., & Lin, J. (2023). Miracl: A multilingual retrieval dataset covering 18 diverse languages. *Transactions of the Association for Computational Linguistics*, 11, 1114–1131.
- Zhang, Y., Kong, X., Shen, Z., Li, J., Yi, Q., Shen, G., & Dong, B. (2024). A survey on temporal knowledge graph embedding: Models and applications. *Knowledge-Based Systems*, 112454.

Zhao, X., Li, M., Lu, W., Weber, C., Lee, J. H., Chu, K., & Wermter, S. (2023). Enhancing zero-shot chain-of-thought reasoning in large language models through logic. *arXiv preprint arXiv:2309.13339*.

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., et al. (2024). Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems*, 36.

Zhong, H., Xiao, C., Tu, C., Zhang, T., Liu, Z., & Sun, M. (2020). Jec-qa: A legal-domain question answering dataset. *Proceedings of the AAAI conference on artificial intelligence*, 34(05), 9701–9708.

Zuccon, G., Koopman, B., & Shaik, R. (2023). Chatgpt hallucinates when attributing answers. *Proceedings of the Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*, 46–51.